transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vials (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma in 2 plastic vials, each containing 0.5 mL

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into separate plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

5. Send specimens in the same shipping container.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85306

PSTF
83049

Protein S Antigen, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). Warfarin will lower protein S. If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vials (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma in 2 plastic vials, each containing 0.5 mL

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot 0.5 mL of plasma into 2 plastic vials, leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

5. Send specimens in the same shipping container.

Additional Information: A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85306-Free

85305-Total (if appropriate)

PST 80994 Protein S Antigen, Total, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see PSTF / Protein S Antigen, Plasma.

Specimen Type: Platelet-poor plasma

Patient Preparation:

1. Patient **should not** be receiving anticoagulant treatment (eg, warfarin, heparin). Warfarin will lower protein S. If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

2. Patient **should not** be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator [tPA]).

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vials (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma in 2 plastic vials, each containing 0.5 mL

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into separate plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

5. Send specimens in the same shipping container.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85305

GNPRS 619187 Protein S Deficiency, PROS1 Gene, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

12PU1 Protein, Total, 12 Hour, Urine

614042

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, 5 mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a 12-hour urine specimen. Specimens should be collected before fluorescein is given or not collected until at least 24 hour later.

2. No preservative.

3. Invert well before taking 4-mL aliquot.

4. Do not overfill aliquot tube, 4 mL at most.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

84156

Protein, Total, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: 24-hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5 mL tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours. Specimens should be collected before fluorescein is given or not collected until at least 24 hour later.

2. No preservative.

3. Invert well before taking 4-mL aliquot.

4. Do not over fill aliquot tube 4 mL at most.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

84156

TPBF
606619

Protein, Total, Body Fluid

Specimen Requirements:

Specimen Type: Body fluid

Preferred Source:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, JP drain)

-Pericardial

Acceptable Source: Write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

84157

PRCON 617782 Protein, Total, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

ORTHP / Orthostatic Proteinuria, Random, Urine

Patient Preparation: Specimens should be collected before fluorescein is given or not collected until at least 24 hour later.

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. This is a nighttime (supine) collection.

2. Void immediately prior to lying down.

3. Collect an 8-hour (nighttime) urine specimen.

4. Upon awaking in the morning, the patient should immediately void into the specimen cup.

4. No preservative.

5. Invert well before taking 4-mL (maximum) aliquot.

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

84156

TP
8520

Protein, Total, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	180 days

CPT Code Information:

84155

TPSF
872

Protein, Total, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge specimen to remove any cellular material.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	72 hours
	Frozen	180 days

CPT Code Information:

84157

RPTU1
614004

Protein/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: Specimens should be collected before fluorescein is given or not collected until at least 24 hour later.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic vial

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

3. Invert well before taking 4 mL aliquot.

4. Do not over fill aliquot tube, 4 mL at most.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

84156

82570

RATO2
617783

Protein/Creatinine, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see ORTHP / Orthostatic Proteinuria, Random, Urine.

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	24 hours

PR3
82965

Proteinase 3 Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

FPT12 Prothrombin Fragment 1+2

75957

Specimen Requirements:

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Do not draw from an arm with a heparin lock or heparinized catheter.

2. Draw blood in a blue-top tube containing 3.2% buffered sodium citrate.

3. Mix immediately by gently inverting at least 6 times.

4. Centrifuge and aliquot 1 mL citrated plasma into a plastic vial.

5. Send frozen.

Critical frozen. Separate specimens must be submitted when multiple tests are ordered.

Specimen Minimum Volume: Plasma: 0.5 mL; Note: This volume does not allow for

repeat testing.

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	28 days

CPT Code Information:

83520

PTNT 81742 Prothrombin G20210A Mutation, Blood

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD solution B), light-blue top (sodium citrate)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube. **Do not** aliquot.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	14 days
	Frozen	14 days
	Refrigerated	14 days

CPT Code Information:

81240-F2 (prothrombin, coagulation factor II) (eg, hereditary hypercoagulability) gene analysis, 20210G>A variant

PTSC 602171 Prothrombin Time (PT), Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile or as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85610

PTMSC 602183 Prothrombin Time Mix 1:1, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85611

PTTP
40934

Prothrombin Time, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL platelet-poor plasma

Collection Instructions:

1. For complete instructions see Coagulation Guidelines for Specimen Handling and Processing

2. Centrifuge, remove plasma, and centrifuge plasma again.

3. Aliquot plasma into plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

Additional Information: A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen (preferred)	30 days
	Ambient	24 hours

CPT Code Information:

85610

PPFWE
31891

Protoporphyrins, Fractionation, Washed Erythrocytes

Specimen Requirements:

Patient Preparation: For 24 hours before specimen collection, patient **must not** consume any alcohol.

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Dark blue top (metal free heparin), green top (lithium heparin), or lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: Entire washed erythrocyte suspension

Collection Instructions:

Collect and process whole blood specimen as follows:

1. Transfer entire specimen to a 12-mL graduated centrifuge tube.

2. Centrifuge specimen at 4 degrees C for 10 minutes at 2000 rpm.

3. Record volume of packed cells and the total volume of the specimen.

4. Discard supernatant plasma.

5. Wash packed erythrocytes 2 times by resuspension of at least an equal amount of cold 0.9% saline, mix, and centrifuge for 5 minutes at 2000 rpm, discarding supernatant after each washing.

6. Resuspend packed cells to the original total volume with 0.9% saline. Invert specimen gently to mix.

7. Transfer washed erythrocytes into a plastic vial and freeze.

Specimen Minimum Volume: Washed and resuspended erythrocytes: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Washed RBC	Frozen (preferred)	14 days
	Refrigerated	14 days

CPT Code Information:

82542

PPFE
8739

Protoporphyrins, Fractionation, Whole Blood

Specimen Requirements:

Patient Preparation: Patient **must not** consume any alcohol for 24 hours before specimen collection.

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA), dark blue top (metal free heparin), royal blue top (metal free EDTA), green top (lithium heparin)

Specimen Volume: 4 mL

Collection Instructions: Refrigerate specimen as soon as possible.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated	7 days

CPT Code Information:

82542

PROTR 9797 Protriptyline (Vivactyl)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable**. Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), **serum gel tube is not acceptable**. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80335

PRSSZ 35532 PRSS1 Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

81404-PRSS1 (protease, serine, 1 [trypsin 1]) (eg, hereditary pancreatitis), full gene sequence

PCHE1 606604 Pseudocholinesterase, Total, Serum

Specimen Requirements:

Patient Preparation: For patients with prolonged apnea following surgery, wait at least 24 hours before obtaining specimen.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	365 days
	Ambient	24 hours

CPT Code Information:

82480

PSY
62235

Psychosine, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Card-Blood Spot Collection (Filter Paper)

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman protein Saver 903 paper, local newborn screening card, or blood collected in tubes containing EDTA (preferred) or heparin and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	96 days	FILTER PAPER
	Frozen	96 days	FILTER PAPER
	Refrigerated	96 days	FILTER PAPER

CPT Code Information:

82542

PSYCF 606146 Psychosine, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial.

Specimen Volume: 0.15 mL

Collection Instructions: Do not aliquot.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen	7 days

CPT Code Information:

82542

PSYR 606145 Psychosine, Whole Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin, lithium heparin)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated	7 days

CPT Code Information:

82542

PSYQP 610060 Psychotropic Pharmacogenomics Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient 4 days (preferred)/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81418

PTFIB
65820

PT-Fibrinogen, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85385

PTNZ
614586

PTEN Hamartoma Tumor Syndrome, PTEN Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81321

FPTH
90182

PTH Antibody

Specimen Requirements:

Specimen Type: Serum

Collection Container/Tube: Red or SST

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and send 1 mL of serum ambient in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Ambient (preferred)	14 days
	Frozen	28 days
	Refrigerated	14 days

CPT Code Information:

83519

PU1
70545

PU.1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

PUSE
82362

Pumpkin Seed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PUPYP
65151

Purines and Pyrimidines Panel, Plasma

Specimen Requirements:

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Plasma

Collection Instructions:

1. Centrifuge at 4 degrees C and aliquot plasma into plastic vial.

2. Send plasma frozen.

Specimen Minimum Volume: Plasma: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	90 days

CPT Code Information:

82542

PUPYU
41977

Purines and Pyrimidines Panel, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative needed.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	90 days

CPT Code Information:

82542

PC1TS
43437

Purkinje Cell Cytoplasmic Antibody Type 1 (PCA-1) Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For further information see:

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

PC1TC
43446

Purkinje Cell Cytoplasmic Antibody Type 1 (PCA-1) Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

PC2TS 43438 Purkinje Cell Cytoplasmic Antibody Type 2 (PCA-2) Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-PAVAL / Paraneoplastic, Autoantibody Evaluation, Serum

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-AIAES / Axonal Neuropathy, Autoimmune/Paraneoplastic Evaluation, Serum

-DYS2 / Dysautonomia, Autoimmune/Paraneoplastic Evaluation, Serum

-GID2 / Gastrointestinal Dysmotility, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

PC2TC
43447

Purkinje Cell Cytoplasmic Antibody Type 2 (PCA-2) Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

PCTTS
43439

Purkinje Cell Cytoplasmic Antibody Type Tr (PCA-Tr) Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-PAVAL / Paraneoplastic Autoantibody Evaluation, Serum

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-PCDES / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

PCTTC 43448 Purkinje Cell Cytoplasmic Antibody, Type Tr (PCA-Tr) Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-PCDEC / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time

CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

FPYRE
57540

Pyrethrum IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

PLP
42359

Pyridoxal 5-Phosphate, Plasma

Specimen Requirements:

Patient Preparation:

1. Fasting: 12 hours, required; Infants should have specimen collected before next feeding, water may be taken as needed

2. For 24 hours before specimen collection, patient **should not take multivitamins or vitamin supplements**.

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Green top (sodium or lithium heparin) or Plasma gel separator (PST)

Acceptable: None

Submission Container/Tube: Amber vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge at 4 degrees C.

2. Aliquot all plasma into amber vial and freeze immediately.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma Heparin	Frozen	29 days	LIGHT PROTECTED

CPT Code Information:

84207

B6PA 42361 Pyridoxic Acid, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see B6PRO / Vitamin B6 Profile (Pyridoxal 5-Phosphate and Pyridoxic Acid), Plasma.

Patient Preparation:

1. **Fasting: 12 hours, required**; Infants-should have specimen collected before next feeding, water can be taken as needed

2. For 24 hours before specimen collection, patient **must not** take multivitamins or

vitamin supplements.

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube: Green top (sodium or lithium heparin) or plasma gel separator (PST)

Submission Container/Tube: Amber vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge at 4 degrees C.

2. Aliquot all plasma into amber vial and freeze immediately.

Specimen Minimum Volume: Plasma: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma Heparin	Frozen	29 days	LIGHT PROTECTED

CPT Code Information:

82542

P5NT
80650

Pyrimidine 5' Nucleotidase, Blood

Specimen Requirements:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 5 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time

Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

83915

PK1 607459 Pyruvate Kinase Enzyme Activity, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

84220

PKC 608418 Pyruvate Kinase Enzyme Activity, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

HAEV1 / Hemolytic Anemia Evaluation, Blood

EEEV1 / Red Blood Cell (RBC) Enzyme Evaluation, Blood

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

84220

PYRC
83356

Pyruvate, Spinal Fluid

Specimen Requirements:

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 0.6 mL

Collection Instructions: Send specimen from vial 4.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	7 days

CPT Code Information:

84210

PYR
8657

Pyruvic Acid, Blood

Specimen Requirements:

Call 800-533-1710 or 507-266-5700 to order special collection tube.

Patient Preparation:

Fasting: 4 hours, required

Supplies: Perchloric Acid-Pyruvate Tube (T012)

Container/Tube: Special collection tube containing 2.5 mL of 6% perchloric acid

Specimen Volume: Exactly 1 mL

Collection Instructions:

1. Special collection tube must be prechilled prior to collection.

2. Draw enough blood directly into syringe to add exactly 1 mL of blood to the prechilled special collection tube.

3. Taking care to not spill any of the preservative, cautiously remove the cap from the tube.

4. Immediately transfer blood, once drawn, to the prechilled, special collection tube, recap, and shake vigorously to mix.

Additional Information:

1. **Check expiration date before using.** Supplied collection tube expires 14 months after preparation.

2. If perchloric acid (preservative) spills, obtain a new, prechilled tube.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated	15 days	PYRUVATE

CPT Code Information:

84210

QFEVR 616832 Q Fever Antibody Screen with Titer Reflex, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

86638

86638 x4 (if appropriate)

QFP
83149

Q Fever IgM and IgG, Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see QFEVR / Q Fever Antibody Screen with Titer Reflex, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

86638 x 4

QUAD1
113145

Quad Screen (Second Trimester) Maternal, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Do not collect specimen after amniocentesis as this could affect results.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Additional Information:

1. For an assessment that includes neural tube defect results, gestational age must be between 15 weeks, 0 days and 22 weeks, 6 days.

2. Assessments for trisomy 21 (Down syndrome) and trisomy 18 (Edwards syndrome) only are available between 14 weeks, 0 days and 22 weeks, 6 days.

3. Initial or repeat testing is determined in the laboratory at the time of report and will be reported accordingly. To be considered a repeat test for the patient, the testing must be within the same pregnancy and trimester, with interpretable results for the same tests, and both tests are performed at Mayo Clinic.

4. Maternal Serum Screening patient education brochure (T522) is available upon request.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

81511

QFT4 113563 QuantiFERON-TB Gold Plus, Blood

Specimen Requirements:

Supplies:

-QuantiFERON-TB Gold Plus Collection Kit (T794)

-QuantiFERON-TB Gold Plus HIGH ALTITUDE Collection Kit (T795)

Collection Instructions:

1. Special collection, incubation, and centrifugation procedures must be followed.

2. For blood collection options (1-tube collection or 4-tube collection) and specimen transport instructions, see Mycobacterium tuberculosis Infection Determination by Quanti-FERON-TB Gold Plus Collection and Processing Instructions (T688).

Specimen Minimum Volume: 4 mL: 1 mL per tube (4 tubes)

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated	28 days	QTBKIT

CPT Code Information:

86480

TBBS 9336 Quantitative Lymphocyte Subsets: T, B, and Natural Killer (NK) Cells, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimen collection be performed at the same time of day.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	52 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86355

86357

86359

86360

QPALM Queen Palm, IgE, Serum
82863

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FQUET Quetiapine (Seroquel)
91727

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube not acceptable. Spin down and send 3 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80342

FQUIN 57922 Quinoa (Chenopodium quinoa) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

REPII Rabbit Epithelium, IgE, Serum
82782

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RAMB Rabbit Meat, IgE, Serum
82860

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RSER
82544

Rabbit Serum Proteins, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RUPR
82148

Rabbit Urine Proteins, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FRFIT
90330

Rabies Antibody Endpoint

Specimen Requirements:

Specimen Type: Serum

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerate.

Note: Collection date is required.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days

CPT Code Information:

86382

FRAD 57933 Radish (Raphanus sativus) IgE

Specimen Requirements:

Specimen Type: Serum

Collection Container/Tube:

Preferred: Red

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

RASE 82366 Rape Seed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RWEED Rape Weed, IgE, Serum

82616

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

BRTP 619957 Rapid Hereditary Breast Cancer Treatment Decision Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81432

MAL 9240 Rapid Malaria/Babesia Smear, Varies

Specimen Requirements:

Both blood and slides are required.

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Type: Blood films

Slides: 2 Thin blood films and 2 thick blood films

Container/Tube: Plastic slide container

Collection Instructions:

1. For complete instructions, see Thick and Thin Blood Slide Preparation Instructions.

2. Slides must be clean and grease-free.

3. Blood films should be made from fresh blood using fingerstick or drops of blood from needle following venipuncture. However, EDTA anticoagulated blood is also acceptable.

4. Prepare thin blood films as follows:

a. Prepare 2 thin smears with a slide prep machine. OR

b. Prepare a thin film with a "feathered edge" that is no more than a single cell thick.

c. Allow the film to thoroughly air dry and then fix by briefly immersing in either absolute or 95% methyl alcohol.

d. Allow to air dry after fixation.

5. Prepare thick blood films as follows:

a. Place a large drop of blood (approximately the size of a dime and preferably from a fingerstick) on a slide.

b. Using a corner of a second slide spread the drop in a circular motion while applying firm pressure to literally scratch the blood onto the carrier slide. This technique allows the blood to dry quickly and adhere well to the slide. Use approximately 20 circular sweeps with the second slide. The drop of blood should be about the size of a quarter when finished.

c. **Do not fix.** Air dry thoroughly (approximately 45 minutes) before placing in transport container.

Specimen Minimum Volume: Blood: 0.5 mL
Slides: See Specimen Required.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

87207

RPRT1 616863 Rapid Plasma Reagin (RPR) Screen with Reflex to Titer, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86592

86593-Rapid Plasma Reagin Titer (if appropriate)

RPRT3 616970 Rapid Plasma Reagin (RPR) with Reflex to Treponema pallidum Particle Agglutination, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86592

86593- (if appropriate)

86780- (if appropriate)

RRPRS 616866 Rapid Plasma Reagin Screen with Reflex to Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see SYPH1 / Syphilis IgG with Reflex, Enzyme Immunoassay, Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Information: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86592-Rapid Plasma Reagin Screen

86593-Rapid Plasma Reagin Titer (if appropriate)

86780-Syphilis Antibody by TP-PA (if appropriate)

RPRS 603261 Rapid Plasma Reagin Screen with Reflex, Serum

Specimen Requirements:

Only available as a reflex test. For more information see SYPHT / Syphilis Total Antibody with Reflex, Serum.

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86593-Rapid Plasma Reagin Titer (if appropriate)

86780-Syphilis Antibody by TP-PA (if appropriate)

RPRT2 616865 Rapid Plasma Reagin Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see RPRT1 / Rapid Plasma Reagin (RPR) Screen with Reflex to Titer, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86593

RPDEI 620377 Rapidly Progressive Dementia Evaluation Interpretation, Spinal Fluid (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
CSF	Frozen (preferred)	28 days	BlueTop SARSTEDT
	Refrigerated	14 days	BlueTop SARSTEDT
	Ambient	12 hours	BlueTop SARSTEDT

RPDE 620376 Rapidly Progressive Dementia Evaluation, Spinal Fluid

Specimen Requirements:

Supplies: CJD/RPD Evaluation Kit (T966)

Container/Tube:

Preferred: 2 Sarstedt CSF False Bottom Tubes 63.614.625 (2.5 mL)

Acceptable: Sarstedt 72.703.600 (1.5 mL) or Sarstedt 72.694.600 (2 mL)

Specimen Volume: 2 tubes; each containing 1.5 to 2.5 mL

Collection Instructions:

1. Perform lumbar puncture and discard the first 1 to 2 mL of cerebrospinal fluid (CSF).

2. Collect two tubes of CSF directly into an acceptable collection tube until the tube is at least 50% full.

3. Send CSF specimen in original collection tube. **Do not aliquot.**

Note: Polystyrene collection tubes are not acceptable. Exposure of CSF to polystyrene tubes may result in falsely low Abeta42 concentrations.

The Alzheimer's Association consensus protocol for handling of CSF for clinical measurements of Abeta42 and tau recommends using the drip method for CSF collection and directly collecting into a low-bind polypropylene tube. Although some clinicians prefer the syringe pull method due to speed of collection, the drip method reduces the risk of Abeta42 binding to the plastic of any syringe used.

4. Collection instructions can also be found on Spinal Fluid Specimen Collection Instructions for Creutzfeldt-Jakob Disease and Rapidly Progressive Dementia Evaluations (T974).

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
CSF	Frozen (preferred)	28 days	BlueTop SARSTEDT
	Refrigerated	14 days	BlueTop SARSTEDT
	Ambient	12 hours	BlueTop SARSTEDT

CPT Code Information:

0584U

82234

84393

84394

RSBV 621403 Rare Subepithelial Autoimmune Blistering Disease Variants, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	14 days

CPT Code Information:

86255

FRASP 57665 Raspberry IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

RASP
86305

Raspberry, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RAT
82725

Rat Epithelium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RTSP
82793

Rat Serum Protein, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RTUP
82794

Rat Urine Protein, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RAVMP
618645

Ravulizumab Monitoring Panel, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, preferred but not required

2. Suggest discontinuing natalizumab at least 4 weeks before specimen collection for ravulizumab testing. Patient should consult the healthcare professional who prescribed this drug to determine if discontinuation is an option. If not, it is okay to proceed with testing while taking natalizumab.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: 2 Plastic vials

Specimen Volume: 2 mL Serum in **2 separate plastic vials** (see Collection Instructions)

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

3. Centrifuge at 4 degrees C.

4. Aliquot serum into two separate plastic vials, each containing 1 mL.

5. Within 30 minutes of centrifugation, freeze specimen. Serum must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the sample is kept on ice before centrifugation, and immediately afterward, the serum is aliquoted and frozen.

Specimen Minimum Volume: Serum: 1 mL in 2 separate plastic vials, each containing 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days
Serum Red	Frozen	14 days

CPT Code Information:

80299

86161

RAVU
609420

Ravulizumab, Serum

Specimen Requirements:

Patient Preparation: Consider discontinuing natalizumab at least 4 weeks prior to specimen collection. Patient should consult the healthcare provider who prescribed this medication to determine if discontinuation is an option. If not, ok to proceed with testing while taking natalizumab.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Immediately after specimen collection, place the tube on wet ice.

3. After specimen has clotted on wet ice, centrifuge at 4 degrees C and aliquot serum into a plastic vial.

4. Freeze specimen within 30 minutes of centrifugation. Specimen must be placed on dry ice if not frozen immediately.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	28 days
	Ambient	28 days
	Refrigerated	28 days

CPT Code Information:

80299

NGSFX 65718 Reanalysis of Acute Myeloid Leukemia 4- or 11- Gene Panels, Additional Genes (Non-Orderable)

Specimen Requirements:

No additional specimen is required. This is a bioinformatics review of additional gene regions not analyzed in the previously ordered NGAMT / MayoComplete Acute Myeloid Leukemia, Therapeutic Gene Mutation Panel (*FLT3*, *IDH1*, *IDH2*, *TP53*), Next-Generation Sequencing, Varies or NGAML / MayoComplete Acute Myeloid Leukemia, 11-Gene Panel, Varies. Call 800-533-1710 for assistance with ordering.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	14 days

CPT Code Information:

81450

NCYB 618977 Recessive Congenital Methemoglobinemia, CYB5 and CYB5 Reductase Genetic Analysis, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

RCVBS 610009 Recoverin-IgG Antibody, Immunoblot, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

84182

EEEV1 Red Blood Cell (RBC) Enzyme Evaluation, Blood
607493

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A)

Specimen Volume: 12 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	11 days

CPT Code Information:

82955-G6PD Enzyme Activity

84087-Glucose Phosphate Isomerase

84220-Pyruvate Kinase Enzyme Activity

82657-Hexokinase

82657-Adenylate Kinase

82657-Phosphofructokinase

82657-Phosphoglycerate Kinase

82657-Triosephosphate Isomerase

82978-Glutathione

83915-Pyrimidine 5' Nucleotidase

EEEVI 608087 Red Blood Cell (RBC) Enzyme Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	11 days

IDCOR 622374 Red Blood Cell Antigen Genotyping, PCR, Blood

Specimen Requirements:

Container/Tube:

Preferred: Pink top (EDTA)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Pediatric Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Label specimen as blood.

3. Send whole blood specimen in the original tube. **Do not aliquot**.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

0084U

NENZ
619047

Red Blood Cell Enzyme Disorders Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies

CPT Code Information:

81443

NMEM 619061 Red Blood Cell Membrane Disorders Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogeneic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

81479

81479 (if appropriate for government payers)

RBCME 64897 Red Blood Cell Membrane Evaluation, Blood

Specimen Requirements:

Both patient and shipping control specimens are required.

For complete instructions, refer to Specimen Collection and Labeling Instructions for Osmotic Fragility Testing of Erythrocytes.

Shipping Control Specimen

Specimen Type: Whole blood (non-patient)

Container/Tube: Lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a shipping control specimen from a healthy person unrelated to the patient at the same time (or within 4 hours) as the patient specimen.

Note: The shipping control specimen can be collected from a phlebotomist, volunteer or another heathy patient.

2. Clearly handwrite "CONTROL" on the outermost label of the tube.

3. Refrigerate (or place on cold gel pack/small amount of wet ice) immediately after collection.

4. Send shipping control in the original tube. **Do not aliquot**.

5. Keep the shipping control and the patient specimens **together**, either rubber banded or in a bag.

Additional Information:

1. **The shipping control and patient specimens must be handled identically** from the time of collection through receipt in the testing laboratory.

2. **If the shipping control is not sent with the patient specimen, test cancellation is likely.**

3. The shipping control specimen evaluates whether the patient result has been compromised by handling conditions such as temperature, motion, or other transportation interferences, as temperature and handling extremes can adversely impact the integrity of the specimen.

Patient Specimen

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. Collect and label patient specimen.

2. Refrigerate (or place on cold gel pack/small amount of wet ice) specimen immediately after collection.

3. Send whole blood specimen in the original tube. **Do not aliquot**.

4. Keep the shipping control and the patient specimens together, either rubber banded or in a bag.

Additional Information: The shipping control and patient specimens must be handled identically from the time of collection through receipt in the testing laboratory.

Patient Specimen

Specimen Type: Slides

Container/Tube: Blood smears

Specimen Volume: 2 Well-made peripheral blood smears

Collection Instructions:

1. Prepare 2 peripheral blood smears from the EDTA tube collected from the patient.

2. Either stain the smear with Wright stain or fix the smear with absolute methanol prior to shipping.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Whole blood patient/Shipping control: 2 mL; Slides: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Control	Refrigerated	72 hours	PURPLE OR PINK TOP/EDTA
Whole Blood EDTA	Refrigerated	72 hours	
Whole Blood Slide	Refrigerated		CARTRIDGE

CPT Code Information:

85557-Osmotic fragility

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

85060-Morphology review

FRSE 57939 Red Snapper (Lutjanus spp) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

SORR 82737

Red Sorrel, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

REDT
82901

Red Top, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

UREDF
607696

Reducing Substance, Feces

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz Random (T288)

Container/Tube: Fecal container

Specimen Volume: 3 g

Collection Instructions:

1. Collect a loose, unpreserved, random fecal specimen.

2. Freeze immediately.

Additional Information: If additional tests are ordered, aliquot and separate sample prior to freezing to allow 1 container per test.

Specimen Minimum Volume: 2 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen	7 days

CPT Code Information:

84376

RBCS
36440

Relative B-Cell Subset Analysis Percentage, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume:

< or =14 years: 4 mL

>14 years: 10 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot.**

2. Label specimen as blood for RBCS.

Additional Information: For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Minimum Volume: < or =14 years: 3 mL; >14 years: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Refrigerated	48 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86356 x7

TFEBF 64973 Renal Cell Carcinoma, 6p21.1 (TFEB) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded (FFPE) tumor tissue block. Blocks prepared with alternative fixation methods will be attempted but are less favorable for successful results by FISH testing.

2. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Bone specimens that have been decalcified will be attempted for testing, but the success rate is approximately 50%.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue and 1 slide stained with hematoxylin and eosin.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), Interpretation and report

88271 x 2-DNA probe, each; each additional probe set (if appropriate)

88271 x 1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

RFAMA 113634 Renal Function Panel, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated	24 hours

CPT Code Information:

KS: 84132

NAS: 84295

CL: 82435

HCO3: 82374

BUN: 84520

CRTS1: 82565

CA: 82310

GLURA: 82947

ALB: 82040

PHOS: 84100

RPCWT 70591 Renal Pathology Consultation, Wet Tissue

Specimen Requirements:

Specimen Type: Kidney biopsy

Supplies: Renal Biopsy Kit (T231)

Specimen Volume: Entire specimen

Collection Instructions: Collect and prepare biopsy specimens per instructions in Renal Biopsy Preparation Instructions.

Additional Information: On Saturdays and holidays, RUSH (same day as receipt) interpretation is available for clinically emergent cases (eg, acute kidney failure, rapidly progressive glomerulonephritis, acute kidney allograft dysfunction) but requires advanced notification and approval by a Mayo Clinic Renal Pathologist (507-284-5677).

To request RUSH service outside of regular business hours, contact Mayo Clinic Laboratories at 800-533-1710.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Kidney Biopsy	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88305 (If appropriate)

88348 (If appropriate)

88313 (If appropriate)

88346 (If appropriate)

88350 (If appropriate)

RFCWT 620415 Renal Pathology Tissue Consultation, Wet Tissue

Specimen Requirements:

Specimen Type: Tissue

Supplies: Renal Biopsy Kit (T968)

Source: Kidney

Specimen Volume: Entire specimen

Collection Instructions: Collect specimens according to the instructions in Renal Biopsy Preparation Instructions.

Additional Information:

Rush (same day as receipt) interpretation is available on Saturday and holidays for clinically emergent cases (eg, acute kidney failure, rapidly progressive glomerulonephritis, acute kidney allograft dysfunction) but requires advanced notification and approval by a Mayo Clinic Renal Pathologist 904-956-3318.

To request RUSH service outside regular business hours, call 800-533-1710.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Kidney Biopsy	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88305 (If appropriate)

88348 (If appropriate)

88313 (If appropriate)

88346 (If appropriate)

88350 (If appropriate)

PRA
8060

Renin Activity, Plasma

Specimen Requirements:

Patient Preparation: For 4 to 6 weeks before specimen collection, spironolactone (Aldactone) should be discontinued, as plasma renin activity cannot be interpreted if the patient is being treated with spironolactone.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Chilled, lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL

Collection Instructions:

1. Draw blood in a chilled syringe from a patient in a seated position; place specimen in a chilled, 3-mL lavender-top (EDTA) tube and mix well. Alternatively, draw blood directly in a chilled, EDTA tube.

2. Immediately place EDTA tube into an ice-water bath until thoroughly cooled.

3. Immediately centrifuge using a refrigerated centrifuge and aliquot plasma into a plastic vial.

Note: If a refrigerated centrifuge is unavailable, chill the centrifuge carriers before use. Centrifuge specimen for 5 minutes or less, then promptly aliquot plasma into a plastic vial.

4. Immediately freeze plasma.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days

CPT Code Information:

84244

RTSCJ
622836

Reptilase Time, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. For complete instruction, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a plastic vial, leaving 0.25 mL in the bottom of centrifuged vial.

4. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally below -40 degrees C.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85635

RTSC
602185

Reptilase Time, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Platelet poor plasma

Collection Instructions:

1. For complete instruction, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85635

RP
609409

Respiratory Panel, PCR, Nasopharyngeal

Specimen Requirements:

Specimen Type: Nasopharyngeal swab

Supplies:

-Culture Swab - Liquid Stuarts/Single Swab (NP Swab) (T515)

-M4-RT (T605)

Collection Container/Tube: Swab. See Additional Information for acceptable swab.

Submission Container/Tube: Transport medium. See Additional Information for acceptable media.

Specimen Volume: Nasopharyngeal swab in minimum volume of 1 mL of transport media

Collection Instructions: Nasopharyngeal swab specimens should be collected according to standard technique and immediately placed into transport media and submitted for testing.

Additional Information:

If any nasopharyngeal swab or transport media not listed below is utilized, testing may be canceled.

-Acceptable nasopharyngeal (NP) swabs are Copan Rayon Swabs, Copan Nylon Flocked Swabs, Copan Polyester Swabs, Puritan Calcium Alginate Swabs, SteriFlock NP Swab

-Acceptable transport media are Remel M4, Remel M4-RT, Remel M5, Remel M6, BD Universal Viral Transport Media (VTM), Copan Universal Transport Media (UTM), PrimeStore Molecular Transport Medium (MTM)

-Acceptable collection and transport system is Sigma-Virocult Viral Collection and Transport System (Swab and transport medium)

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	72 hours
	Frozen	30 days
	Ambient	4 hours

CPT Code Information:

0202U

RPB 616389 Respiratory Panel, PCR, Varies

Specimen Requirements:

Specimen Type: Respiratory fluid

Sources: Bronchoalveolar lavage (BAL) or bronchial washing

Collection Container/Tube: Sterile container

Specimen Volume: 1 mL

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	72 hours
	Frozen	30 days
	Ambient	4 hours

CPT Code Information:

87633

RPR1 62046 Respiratory Profile, Region 1, North Atlantic (CT, MA, ME, NJ, NH, NY, PA, RI, VT), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.8 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785

86003 x 25

RPR10 62056 Respiratory Profile, Region 10, Southwestern Grasslands (OK, TX), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.55 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 25 Each individual allergen

RPR11 62057 Respiratory Profile, Region 11, Rocky Mountain (AZ [Mt]; CO; ID [Mt]; NM, UT [Mt]; WY), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 24-Each individual allergen

RPR12 62058 Respiratory Profile, Region 12, Arid Southwest (Southern AZ Desert, Southern CA Desert), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 22-Each individual allergen

RPR13 62059 Respiratory Profile, Region 13, Southern Coastal California, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 24-Each individual allergen

RPR14 62060 Respiratory Profile, Region 14, Central California, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 23-Each individual allergen

RPR15 62061 Respiratory Profile, Region 15, Intermountain West (Southern ID, NV), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 22-Each individual allergen

RPR16 62062 Respiratory Profile, Region 16, Inland Northwest (OR, Central and Eastern WA), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 21-Each individual allergen

RPR17 62063 Respiratory Profile, Region 17, Pacific Northwest (Northwestern CA, Western OR, WA), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 23-Each individual allergen

RPR18 62064 Respiratory Profile, Region 18, Alaska, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 15-Each individual allergen

RPR19 Respiratory Profile, Region 19, Puerto Rico, Serum

62065

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 21-Each individual allergen

RPR2 62047 Respiratory Profile, Region 2, Mid-Atlantic (DC, DE, MD, NC, VA), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 23-Each individual allergen

RPR3 62048 Respiratory Profile, Region 3, South Atlantic (GA, N.FA, SC), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 22-Each individual allergen

RPR4 62049 Respiratory Profile, Region 4, Subtropic Florida (Florida S. of Orlando), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 22-Each individual allergen

RPR5 62050 Respiratory Profile, Region 5, Ohio Valley (IN, KY, OH, TN, WV), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 26-Each individual allergen

RPR6 62051 Respiratory Profile, Region 6, South Central (AL, AR, LA, MS), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 22-Each individual allergen

RPR7 62052 Respiratory Profile, Region 7, Northern Midwest (MI, MN, WI), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 23-Each individual allergen

RPR8 62053 Respiratory Profile, Region 8, Central Midwest (IA, IL, MO), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.55 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 25-Each individual allergen

RPR9 62054 Respiratory Profile, Region 9, Great Plains (KS, ND, NE, SD), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.6 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

82785-IgE

86003 x 22-Each individual allergen

RSVAB 601948 Respiratory Syncytial Virus (RSV) In Situ Hybridization, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-4 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88365-TC, Primary

88364-TC, if additional ISH

FRSVQ 75669 Respiratory Syncytial Virus (RSV) RNA, Qualitative Real-Time PCR

Specimen Requirements:

Specimen Type: Bronchoalveolar lavage or bronchial wash

Container/Tube: Sterile container

Specimen Volume: 2 mL

Collection Instructions:

Collect in a sterile leak-proof container (no media or preservative). Ship refrigerated.

Note: Specimen type is required.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	30 days

	Ambient	7 days

CPT Code Information:

87634

RETB
616856

Reticulocyte Profile, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	48 hours
	Ambient	24 hours

CPT Code Information:

85046

RB1
604028

Retinoblastoma Protein (Rb) Immunostain, Tech Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained

slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FRBP
75570

Retinol Binding Protein

Specimen Requirements:

Specimen Type: Serum

Patient Preparation:

Fasting: 12 hours, required

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Specimen Volume: 2 mL serum

Collection Instructions:

1. Centrifuge and aliquot 2 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	20 days
	Frozen	180 days

CPT Code Information:

83883

RB24
609449

Retinol-Binding Protein, 24 Hour, Urine

Specimen Requirements:

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. No preservative.

3. Mix well before taking 5-mL aliquot.

Additional Information:

See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83883

RBR
610010

Retinol-Binding Protein, Random, Urine

Specimen Requirements:

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83883-Retinol -Binding Protein, Random, U

82570-Creatinine, Random, U

RAPAN
616785

Rheumatoid Arthritis Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86200

86431

RFPN
621421

Rheumatoid Factor Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	48 hours

CPT Code Information:

86431 x2

RHUT
603415

Rheumatoid Factor, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

86431

RHNI
82856

Rhizopus nigricans, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FRDG
57959

Rhodotorula IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FRUB 57920 Rhubarb (Rheum rhaponticum) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

VITB2
42363

Riboflavin (Vitamin B2), Plasma

Specimen Requirements:

Patient Preparation:

Fasting: 12 hours, required; Infants should have specimen collected before next feeding

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Green top (sodium or lithium heparin)

Acceptable: Light-green top (sodium or lithium heparin plasma gel)

Submission Container/Tube: Amber vial

Specimen Volume: 2 mL Plasma

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into a amber vial.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Plasma Heparin	Refrigerated (preferred)	28 days	LIGHT PROTECTED
	Frozen	28 days	LIGHT PROTECTED
	Ambient	72 hours	LIGHT PROTECTED

CPT Code Information:

84252

RIB
87837

Ribosome P Antibodies, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

FRICE
57633

Rice IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

RICE
82709

Rice, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FMTAG
57260

Rickettsia typhi (murine typhus) Antibodies, IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	90 days
	Refrigerated	5 days

CPT Code Information:

86757

FRIFA
75628

Rifampin Level

Specimen Requirements:

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Allow serum to clot for 30 minutes.

2. Immediately centrifuge and aliquot 2 mL of serum into a plastic vial.

3. Send frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	365 days

CPT Code Information:

80299

RISAP
621813

Risankizumab Quantitation with Antibodies, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose (trough specimen).

2. Within 2 hours of collection, centrifuge, and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

80299

82397

RISA
621304

Risankizumab, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Draw blood immediately before the next scheduled dose (trough specimen).

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

ROMA2
46917

Risk Score, if Postmenopausal, Serum (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	84 days

	Refrigerated	48 hours

ROMA1 Risk Score, if Premenopausal, Serum (Non-Orderable)

46916

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	84 days
	Refrigerated	48 hours

FRISP Risperidone (Risperdal) and 9-Hydroxyrisperidone

91105

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Specimen Type: Plasma (Preferred)

Container/Tube: Green-top (sodium heparin) tube(s).

Specimen volume: 3 mL

Collection Instructions: Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 3 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Specimen Type: Serum

Container/Tube: Red-top tube, serum gel is not acceptable.

Specimen volume: 3 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80342

RIVAR
65847

Rivaroxaban, Anti-Xa, Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. Specimen should be collected 2 to 4 hours (peak) after a dose or just prior (trough) to the next dose for rivaroxaban concentrations.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma into a plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	42 days

CPT Code Information: 80299

RNAP
83397

RNA Polymerase III Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

RNP
81357

RNP Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235

ROPAN
615248

Ro52 and Ro60 Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	21 days

CPT Code Information:

86235 x 2

RO52
615246

Ro52 Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	21 days

CPT Code Information:

86235

RO60
615247

Ro60 Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	21 days

CPT Code Information:

86235

ROMA
62661

ROMA Score (Ovarian Malignancy Risk Algorithm), Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	84 days
	Refrigerated	48 hours

CPT Code Information:

86305-HE4, S

86304-Cancer Ag 125 (CA 125), S

FROPI
57171

Ropivacaine, Serum/Plasma

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Submit only 1 of the following specimens:

Serum

Draw blood in a plain, red-top tube(s). (Serum gel tube is not acceptable.) Spin down and send 1 mL serum refrigerate in a plastic vial.

Plasma

Draw blood in a lavender-top (EDTA) or pink top tube(s). (Plasma gel tube is not acceptable.) Spin down and send 1 mL plasma refrigerate in plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	30 days
	Frozen	90 days
	Ambient	30 days

CPT Code Information:

80299

ROTA
8886

Rotavirus Antigen, Feces

Specimen Requirements:

Specimen Type: Fresh unpreserved feces

Supplies: Stool Collection Kit, Random (T635)

Container/Tube:

Preferred: Sterile fecal container

Acceptable: Swab

Specimen Volume: 5 to 10 grams

Collection Instructions: Place specimen in a tightly sealed plastic bag.

Specimen Minimum Volume: 1 gram

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	7 days
	Refrigerated	72 hours

CPT Code Information:

87425

MARS
82701

Rough Marsh Elder, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RRRP
82723

Rough Pigweed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RBPG
34938

Rubella Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Instructions: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86762

ROC
5194

Rubeola (Measles) Antibodies, IgG and IgM, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 0.25 mL

Collection Instructions: Submit specimen from collection vial number 2 (preferred), 3, or 4.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86765 x 2

RUFI
63030

Rufinamide, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Collect blood immediately before next scheduled dose.

2. For sustained-release formulations ONLY, collect blood a minimum of 12 hours after last dose.

3. Within 2 hours of collection, centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80210

T821Q 615905 RUNX1::RUNX1T1 Translocation (8;21), Minimal Residual Disease Monitoring, Quantitative, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Label specimen as blood.

Specimen Type: Bone marrow aspirate

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Minimum Volume: Peripheral blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	5 days
	Ambient	72 hours

CPT Code Information:

81401

RUSS
82681

Russian Thistle, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FRFYG
57579

Rye Food IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

RYEG
82908

Rye Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RYE
82689

Rye, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

2SC
610031

S-(2-Succinyl)-Cysteine (2SC) Immunostain, Technical Component Only

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostain Technical Only Envelope

Preferred: 2 unstained positively charged glass slide (25- x 75- x 1-mm) per test ordered; sections 4-microns thick.

Acceptable: Formalin-fixed, paraffin-embedded (FFPE) tissue block

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

S100
70547

S-100 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

F100B
57349

S-100B Protein, Serum

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL serum

Collection Instructions:

1. Allow specimen to clot completely at ambient temperature.

2. Within 2 hours of collection, centrifuge and aliquot 2 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	24 hours

CPT Code Information:

86316

SSCTU 607001 S-Sulfocysteine Panel, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative needed.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	90 days

CPT Code Information:

82542

SCERA 610002 Saccharomyces cerevisiae Antibody, IgA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86671

SCERG
610003

Saccharomyces cerevisiae Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86671

FSFLE
57541

Safflower (Carthamus tinctorius) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

FSAG 57957 Sage (Artemisia spp.) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

SALCA Salicylate, Serum
37061

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80179

SALL4 SALL4 Immunostain, Technical Component Only
71534

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FSALG
57631

Salmon IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.
2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.
3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

SALM
82754

Salmon, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SALMC
606220

Salmonella Culture, Feces

Specimen Requirements:

Patient Preparation: Patient **should not** use barium or bismuth for 7 to 10 days before specimen collection.

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of fecal specimen

Collection Instructions:

1. Collect fresh feces and submit 1 gram or 5 mL in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

3. Place vial in a sealed plastic bag.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

87046-Salmonella Culture, Stool-with isolation and preliminary examination

87077-Bacteria Identification (if appropriate)

87153-Aerobe Ident by Sequencing (if appropriate)

87077-Additional Identification Procedure (if appropriate)

87147 x 1-3-Serologic Agglut Method 1 Ident (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

HEXBZ
608026

Sandhoff Disease, HEXB Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes

prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies

CPT Code Information:

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

SARD
82818

Sardine (Pilchard), IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SATB2
607600

SATB2 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

C5B9
616921

SC5b-9 Level Terminal Complement Complex, Plasma

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, preferred but not required

2. **Do not** collect specimens for at least 48 hours following plasma exchange.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Lavender top (K2 EDTA)

Acceptable: Lavender top (K3 EDTA), light-blue top (sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Plasma

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice.

2. Centrifuge between 1000 and 2000 x *g* for 10 minutes at 4 degrees C.

3. Aliquot plasma into a plastic vial.

4. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the sample is kept on ice before centrifugation, and immediately afterward, the plasma is aliquoted and frozen.

Specimen Minimum Volume: Plasma: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days

CPT Code Information:

86160

SCLE
82716

Scale, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SCALS
82259

Scallop, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SHUR
60451

Schistosoma Exam, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Collect a random urine specimen. Preferred time of collection between the hours of

12 noon and 3 p.m. but not required. A 24-hour urine collection is also acceptable.

2. No preservative.

Additional Information: A specimen volume of at least 10 mL is recommended for optimal recovery of *Schistosoma* species ova.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated	7 days

CPT Code Information:

87210

87015

BILHA
65019

Schistosoma species Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86682

SCL70
80178

Scl 70 Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235

FSCPR
75625

Scleroderma Comprehensive Profile

Specimen Requirements:

Collection Container/Tube: 10 mL Red

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Acceptable: SST

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 1 mL (volume does not allow for repeat testing)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

86038

83516

86235 x 5

86256

FSCN4
75263

SCN4A (Myotonia) DNA Sequencing Test

Specimen Requirements:

Note: Collection date and informed consent are required.

Container/Tube: Lavender-top (EDTA)

Specimen Volume: 8 mL

Pediatric (0-3 years) **Volume**: 2 mL

Collection Instructions:

1. Invert several times to mix.

2. Send ambient.

Additional Information: Higher blood volumes ensure adequate DNA quantity, which varies with WBC, specimen condition, and need for confirmatory testing. Patients aged 0 to 3 years have higher WBC, yielding more DNA per mL of blood.

Specimen Minimum Volume: 6 mL; Pediatric (0-3 years): 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Ambient (preferred)	10 days
	Refrigerated	10 days

CPT Code Information:

81406

SDHMP 619712 SDH Genes Mutation Analysis, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

15 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81404

81405 x2

81406

81479 (if appropriate for government payers)

SDHB
70550

SDHB Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SEAFP
31770

Seafood Allergen Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 5

SEAS
31766

Seasonal Inhalants Allergen Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 10

SECOS
8243

Secobarbital, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Collect specimen immediately before the next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

FSHPU
58038

Sedative Hypnotic Panel, Urine

Specimen Requirements:

Collect 10 mL random urine without preservatives. Ship refrigerated in a plastic container.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80307

SEWB
65600

Selenium, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for at least 96 hours.**

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 0.8 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time

Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

84255

SES 9765 Selenium, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium, iodine, and barium are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium-, iodine-, and barium containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies:

-Metal Free Specimen Vial (T173)

-Metal Free B-D Tube (No Additive), 6 mL (T184)

Collection Container/Tube: 6-mL Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Metal-free, screw-capped, polypropylene vial

Specimen Volume: 0.8 mL serum

Collection Instructions:

1. Allow the specimen to clot for 30 minutes; then centrifuge the specimen to separate serum from the cellular fraction.

2. Remove the stopper. Carefully pour specimen into a metal-free, polypropylene vial, avoid transferring the cellular components of blood. **Do not** insert a pipette into the serum to accomplish transfer, and **do not** ream the specimen with a wooden stick to assist with serum transfer.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	28 days	METAL FREE
	Ambient	28 days	METAL FREE

	Frozen	28 days	METAL FREE

CPT Code Information:

84255

SEMA3 617055 Semaphorin 3B (SEMA3B) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SEMB 60556 Semen Analysis with Strict Morphology, Semen

Specimen Requirements:

Patient Preparation: Patient should have 2 to 7 days of sexual abstinence at the time of semen collection for accurate results.

Supplies: Semen Analysis Kit - Dilution Media (T178)

Specimen Volume: Total ejaculate

Collection Instructions:

See Semen Collection Specimen Preparation and Packaging Instructions included with the kit

1. Prior to use, allow dilution medium to warm to room temperature for 45 to 60 minutes.

2. Allow semen to liquefy at room temperature for up to 30 minutes.

3. Use sterile volumetric pipet or tube for volume measurement.

4. Pour liquefied semen into 50-mL dilution medium container within 60 minutes of semen collection time, cap tightly, but do not overtighten, and gently mix.

Note: Proper temperature maintenance of specimen throughout processing and shipping is critical. All materials the specimen is exposed to should be at room temperature (20–28 degrees C).

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Semen	Ambient	36 hours

CPT Code Information:

89310-Semen Analysis

89398-Strict Criteria Sperm Morphology

If both components performed,

89322-Semen Analysis with Strict Morphology

FER
81641

Semen Analysis, Semen

Specimen Requirements:

Patient Preparation: Patient should have 2 to 7 days of sexual abstinence at the time of semen collection for accurate results.

Supplies: Semen Analysis Kit - Dilution Media (T178)

Specimen Volume: Total ejaculate

Collection Instructions:

See complete Semen Collection instructions included with the kit.

1. Prior to use, allow dilution medium to warm to room temperature for 45 to 60 minutes.

2. Allow semen to liquefy at room temperature for up to 30 minutes.

3. Use sterile volumetric pipet or tube for volume measurement.

4. Pour liquefied semen into 50-mL dilution medium container within 60 minutes of semen collection time, cap tightly, but do not overtighten, and gently mix.

Note: Proper temperature maintenance of specimen throughout processing and shipping is critical. All materials the specimen is exposed to should be at room temperature (20-28 degrees C).

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Semen	Ambient	

CPT Code Information:

89310

SMFL
82858

Seminal Fluid, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL

More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SP5CS 615869 Septin-5 Antibody, Cell Binding Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

SP5CC 615868 Septin-5 Antibody, Cell-Binding Assay, Spinal Fluid (Non-

Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid.

Container/Tube: Sterile vial

Specimen Volume: 1.5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

SP5TS 616113 Septin-5 Antibody, Tissue Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

SP5TC
616114

Septin-5 Antibody, Tissue Immunofluorescence, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid.

Container/Tube: Sterile vial

Specimen Volume: 1.5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

SP7CS
615873

Septin-7 Antibody, Cell-Binding Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

SP7CC 615872 Septin-7 Antibody, Cell-Binding Assay, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-ENC2 / Encephalopathy, Autoimmune /Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune /Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 1.5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

SP7TS 616115 Septin-7 Antibody, Tissue Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

SP7TC 616116 Septin-7 Antibody, Tissue Immunofluorescence Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-ENC2 / Encephalopathy, Autoimmune /Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune /Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Autoimmune Myelopathy /Paraneoplastic Evaluation, Spinal Fluid

Container/Tube: Sterile vial

Specimen Volume: 1.5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

SEP9Z
617714

SEPTIN9 Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

FSERO
75621

SeroNeg RAdx3 Profile

Specimen Requirements:

Collection Container/Tube: 10 mL Red

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Acceptable: SST

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is acceptable. Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 3 mL (volume does NOT allow for repeat testing)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	60 days
	Ambient	7 days

CPT Code Information:

83520

83516 x2

SRAU
616216

Serotonin Release Assay, Unfractionated Heparin, Mass Spectrometry, Serum

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, preferred but not required; Collect during an episode of suspected heparin-induced thrombocytopenia

2. Patient **should not** be on ticagrelor (Brilinta) as this may interfere with the assay, yielding a false-negative result.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not** acceptable)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Serum

Collection Instructions:

1. After collection, allow the specimen to sit for a minimum of 1 hour at ambient temperature to ensure it clots completely.

2. Within 48 hours of collection, centrifuge and aliquot serum into a 13-mm x 75-mm, 5-mL, polypropylene vial

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen (preferred)	
	Refrigerated	7 days

CPT Code Information:

82542

SERU
87834

Serotonin, 24 Hour, Urine

Specimen Requirements:

Patient Preparation:

1. For 48 hours before and during specimen collection, patient should **not** eat avocados, bananas, butternuts, cantaloupe, dates, eggplant, grapefruit, hickory nuts, honeydew melon, kiwifruit, melon, nuts, pecans, pineapple, plantains, plums, tomatoes, or walnuts, which are high in serotonin or tryptophan.

2. Patient **should not** take medications that may elevate serotonin levels, including lithium, monoamine oxidase inhibitors, methyldopa, morphine, and reserpine, or selective serotonin reuptake inhibitors (SSRI, eg, PROZAC) which can lead to depletion of platelet serotonin levels and result in false-negative serotonin results for a minimum of 72 hours before specimen collection. Some drugs with longer half-lives (i.e. fluoxetine) can require months after discontinuation for serotonin levels to return to baseline.

3. Patient should avoid heavy nicotine consumption during the 24-hour collection period.

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Add 25 mL of 50% acetic acid as preservative at start of collection.

2. Collect urine for a full 24 hours (required) and record the total volume.

3. Refrigerate specimen during collection.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	48 hours

CPT Code Information:

84260

SERWB Serotonin, Blood

84373

Specimen Requirements:

Patient Preparation: Patient **should not** take medications that may elevate serotonin levels, including lithium, monoamine oxidase inhibitors, methyldopa, morphine, and reserpine, or selective serotonin reuptake inhibitors (SSRI, eg, PROZAC) which can lead to depletion of platelet serotonin levels and result in false-negative serotonin results for a minimum of 72 hours before specimen collection. Some drugs with longer half-lives (i.e. fluoxetine) can require months after discontinuation for serotonin levels to return to baseline.

Supplies: Serotonin Tube (T259)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Serotonin tube containing ascorbic acid

Specimen Volume: 2.5 mL

Collection Instructions:

1. Immediately after collection, transfer approximately 2.5 mL of whole blood to the serotonin tube and mix well (any volume of whole blood from 1.5-3 mL is acceptable).

2. Immediately freeze specimen (necessary to lyse the red blood cells).

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Frozen	90 days	SEROTONIN TUBE

CPT Code Information:

84260

SER
84395

Serotonin, Serum

Specimen Requirements:

Patient Preparation: Patient **should not** take medications that may elevate serotonin levels, including lithium, monoamine oxidase inhibitors, methyldopa, morphine, and reserpine, or selective serotonin reuptake inhibitors (SSRI, eg, PROZAC) which can lead to depletion of platelet serotonin levels and result in false-negative serotonin results for a minimum of 72 hours before specimen collection. Some drugs with longer half-lives (i.e. fluoxetine) can require months after discontinuation for serotonin levels to return to baseline.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL

Collection Instructions: Centrifuge as soon as blood has clotted and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	90 days
	Ambient	4 days

CPT Code Information:

84260

SERPZ
63128

SERPINA1 Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days /Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

FSERT
91345

Sertraline (Zoloft) and Desmethylsertraline

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Specimen Type: Plasma

Container/Tube: Green Top

Specimen Volume: 2 mL

Collection Instructions: Draw blood in a green-top (sodium heparin) tube, **plasma gel tube is not acceptable.** Spin down and send 2 mL of sodium heparin plasma refrigerated in a plastic vial.

Serum

Specimen Type: Serum

Container/Tube: Red

Specimen Volume: 2 mL

Collection Instructions: Draw blood in a plain, red-top tube, **serum gel tube is not acceptable.** Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days

	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80332

FSESG 57682 Sesame Seed IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

SESA 82728 Sesame Seed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SESPF 621157 Sesame Seed, IgE, with Reflex to Sesame Seed Component, IgE, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SCOF3 622027 Severe Acute Respiratory Syndrome Coronavirus 2 (SARS CoV-2) and Influenza Virus Type A and Type B RNA, Molecular Detection, PCR, Varies

Specimen Requirements:

Preferred:

Specimen Type: Nasopharyngeal swab

Container/Tube: Sterile container with viral transport media

Specimen Volume: Entire specimen with a minimum of 1.5 mL (maximum 3 mL) of transport media.

Collection Instructions:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Swab must be placed into viral transport media (eg, M4-RT, M4 or M5), saline, or phosphate buffered saline (PBS). Media **should not** contain guanidine thiocyanate (GTC).

Specimen Type: Bronchoalveolar lavage fluid

Container/Tube: Sterile container

Specimen Volume: 0.6 mL

Additional Information: Do not aliquot into viral transport media.

Acceptable:

Specimen Types: Oropharyngeal (throat) swab, nasal mid-turbinate, or nares/nasal swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-Swab, Sterile Polyester (T507)

Collection Container/Tube:

Preferred: BBL Culture Swab, COPAN Mid-turbinate Swab

Acceptable: Dacron-tipped swab with plastic handle

Submission Container/Tube: Sterile container with transport media

Specimen Volume: Entire specimen with a minimum of 1.5 mL (maximum 3 mL) of transport media.

Collection Instructions: Swab must be placed into viral transport media (eg, M4-RT, M4, or M5), saline, or PBS. Media **should not** contain guanidine thiocyanate (GTC).

Specimen Types: Bronchial washings, endotracheal aspirate, sputum

Container/Tube: Sterile container

Specimen Volume: 0.6 mL

Additional Information: Do not aliquot into viral transport media.

Specimen Minimum Volume: Upper respiratory tract swab: See Specimen Required; lower respiratory specimens: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen (preferred)	14 days
	Refrigerated	72 hours

CPT Code Information:

87636

SFRB 617492 Severe Acute Respiratory Syndrome Coronavirus 2 (SARS CoV-2), Influenza A and B, and Respiratory Syncytial Virus, Molecular Detection, Bronchoalveolar Lavage

Specimen Requirements:

Source: Bronchoalveolar lavage (BAL)

Container/Tube: Sterile container

Specimen Volume: 1 mL

Additional Information: Do not aliquot into viral transport medium.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Bronchoalveolar Lavage	Refrigerated (preferred)	72 hours
	Ambient	24 hours

CPT Code Information:

87637

SCOF4
622031

Severe Acute Respiratory Syndrome Coronavirus 2 (SARS CoV-2), Influenza Virus Type A and Type B RNA, and Respiratory Syncytial Virus (RSV) Molecular Detection, PCR, Varies

Specimen Requirements:

Preferred:

Specimen Type: Nasopharyngeal swab

Container/Tube: Sterile container with transport media

Specimen Volume: Entire specimen with a minimum of 1.5 mL (maximum 3 mL) of transport media.

Collection Instructions:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Swab must be placed into viral transport media (eg, M4-RT, M4 or M5), saline, or phosphate buffered saline (PBS). Media **should not** contain guanidine thiocyanate (GTC).

Specimen Type: Bronchoalveolar lavage fluid

Container/Tube: Sterile container

Specimen Volume: 0.6 mL

Additional Information: Do not aliquot into viral transport media.

Acceptable:

Specimen Type: Oropharyngeal (throat) swab, nasal mid-turbinate, or nares/nasal swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-Swab, Sterile Polyester (T507)

Collection Container/Tube:

Preferred: BBL Culture Swab, COPAN Mid-turbinate Swab

Acceptable: Dacron-tipped swab with plastic handle

Submission Container/Tube: Sterile container with transport media

Specimen Volume: Entire specimen with a minimum of 1.5 mL (maximum 3 mL) of transport media.

Collection Instructions: Swab must be placed into viral transport media (eg, M4-RT, M4, or M5), saline, or PBS. Media **should not** contain guanidine thiocyanate (GTC).

Specimen Type: Bronchial washings, endotracheal aspirate, sputum

Container/Tube: Sterile container

Specimen Volume: 0.6 mL

Additional Information: Do not aliquot into viral transport media.

Specimen Minimum Volume: Upper respiratory tract swab: See Specimen Required; lower respiratory specimens: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen (preferred)	14 days
	Refrigerated	72 hours

CPT Code Information:

87637

CoV-2), Molecular Detection, Varies

Specimen Requirements:

Preferred:

Specimen Type: Nasopharyngeal swab

Container/Tube: Sterile container with viral transport media

Specimen Volume: Entire specimen with a minimum of 1.5 mL (maximum 3 mL) of transport media.

Collection Instructions:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Swab must be placed into viral transport media (eg, M4-RT, M4 or M5), saline, or phosphate buffered saline (PBS). Media **should not** contain guanidine thiocyanate (GTC).

Specimen Type: Bronchoalveolar lavage fluid

Container/Tube: Sterile container

Specimen Volume: 0.6 mL

Additional Information: **Do not aliquot into viral transport media.**

Acceptable:

Specimen Type: Oropharyngeal (throat) swab, nasal mid-turbinate, or nares/nasal swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-Swab, Sterile Polyester (T507)

Collection Container/Tube:

Preferred: BBL Culture Swab or COPAN Mid-turbinate Swab

Acceptable: Dacron-tipped swab with plastic handle

Submission Container/Tube: Swab in sterile container with transport media

Specimen Volume: Entire specimen with a minimum of 1.5 mL (maximum 3 mL) of

transport media.

Collection Instructions: Swab must be placed into viral transport media (eg, M4-RT, M4, or M5), saline, or PBS. Media **should not** contain guanidine thiocyanate (GTC).

Specimen Type: Bronchial washings, endotracheal aspirate, sputum

Container/Tube: Sterile container

Specimen Volume: 0.6 mL

Additional Information: Do not aliquot into viral transport media.

Specimen Minimum Volume: Upper respiratory tract swab: See Specimen Required; lower respiratory specimens: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen (preferred)	14 days
	Refrigerated	72 hours

CPT Code Information:

87635

COVTA 622465 Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), Nucleocapsid, Total Antibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

86769

COVSQ 622462 Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), Spike Antibody, Semi-Quantitative, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86769

SCIDP 620134 Severe Combined Immunodeficiency (SCID) Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen **will not be accepted**.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Label specimen as bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerate 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233- Tissue culture, skin, solid tissue biopsy (if appropriate)

88240- Cryopreservation (if appropriate)

SCCNP 619872 Severe Congenital and Cyclic Neutropenia Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred)/Refrigerated (<24 hours)

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Minimum Volume: Blood: 1 mL; Skin biopsy or cultured fibroblasts: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233- Tissue culture, skin, solid tissue biopsy (if appropriate)

88240- Cryopreservation (if appropriate)

SCTF 35843 Sex Chromosome Determination, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), Interpretation and report

88271 x 2-DNA probe, each; each additional probe set (if appropriate)

88271-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

SHBG1 Sex Hormone-Binding Globulin, Serum

608102

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

84270

SRYF
35301

Sex-Determining Region Y, Yp11.3 Deletion, FISH

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 to 25 mL

Collection Instructions:

1. Optimal timing for specimen collection is during 14 to 18 weeks of gestation, but specimens collected at other weeks of gestation are also accepted. Provide gestational age at the time of amniocentesis.

2. Discard the first 2 mL of amniotic fluid.

3. If ordering with CMAP / Chromosomal Microarray, Prenatal, Amniotic

Fluid/Chorionic Villus Sampling, submit a minimum of 12 mL.

4. If ordering with CHRAF / Chromosome Analysis, Amniotic Fluid, submit a minimum of 12 mL.

5. If ordering with both CMAP and CHRAF, submit a minimum of 26 mL.

Additional Information:

1. Unavoidably, about 1% to 2% of mailed-in specimens are not viable.

2. Bloody specimens are undesirable.

3. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

4. Specimen cannot be frozen.

Specimen Type: Autopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 4 mm diameter

Collection Instructions:

1. Wash biopsy site with an antiseptic soap.

2. Thoroughly rinse area with sterile water.

3. Do not use alcohol or iodine preparations.

4. Biopsy specimens are best taken by punch biopsy to include full thickness of dermis.

Specimen Type: Whole blood

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Other anticoagulants are not recommended and are harmful to the viability of the cells.

4. Cord blood is acceptable

Additional Information:

1. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

2. Specimen cannot be frozen.

Specimen Type: Chorionic villus

Supplies: CVS Media (RPMI) and Small Dish (T095)

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 to 25 mg

Collection Instructions:

1. Collect specimen by the transabdominal or transcervical method.

2. Transfer chorionic villi to a Petri dish containing transport medium.

3. Using a stereomicroscope and sterile forceps, assess the quality and quantity of the villi and remove any blood clots and maternal decidua.

Specimen Type: Fixed cell pellet

Container/Tube: Sterile container with a 3:1 fixative (methanol:glacial acetic acid)

Specimen Volume: Entire specimen

Specimen Type: Products of conception or stillbirth

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 1 cm(3) of placenta (including 20-mg of chorionic villi) **and** a 1-cm(3) biopsy specimen of muscle/fascia from the thigh

Collection Instructions: If a fetus cannot be specifically identified, collect villus material or tissue that appears to be of fetal origin.

Additional Information: Do not send entire fetus.

Specimen Type: Skin biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 4 mm diameter

Collection Instructions:

1. Wash biopsy site with an antiseptic soap.

2. Thoroughly rinse area with sterile water.

3. **Do not** use alcohol or iodine preparations.

4. A local anesthetic may be used.

5. Biopsy specimens are best taken by punch biopsy to include full thickness of dermis.

Specimen Minimum Volume: Amniotic fluid: 5 mL; Autopsy, skin biopsy: 4 mm; Whole blood: 2 mL; Chorionic villi: 5 mg; Fixed cell pellet: 1 pellet

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88273 w/modifier 52-Chromosomal in situ hybridization, less than 10 cells (if appropriate)

88273-Chromosomal in situ hybridization, 10-30 cells (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

SZDIA
64750

Sezary Diagnostic Flow Cytometry, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA), green top (sodium heparin)

Specimen Volume: 6 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot**.

2. Label specimen as blood.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88188-Flow Cytometry Interpretation, 9 to15 markers (if appropriate)

88189-Flow Cytometry Interpretation, 16 or more markers (if appropriate)

SZMON
64749

Sezary Monitoring Flow Cytometry, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA), green top (sodium heparin)

Specimen Volume: 6 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot**.

2. Label specimen as blood.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88188-Flow Cytometry Interpretation, 9 to15 markers (if appropriate)

88189-Flow Cytometry Interpretation, 16 or more markers (if appropriate)

SF1
72121

SF-1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SHWL
82747

Sheep Wool, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

STFRP
35148

Shiga Toxin, Molecular Detection, PCR, Feces

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen

to be processed in an environment in which contamination of the specimen by shiga toxin DNA is unlikely.

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Preserved feces

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of nonnutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of feces; 5 mL

Collection Instructions:

1. Collect fresh fecal specimen and submit in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

Specimen Stability Information: Ambient (preferred) 7 days/Refrigerated 7 days

Acceptable:

Specimen Type: Unpreserved feces

Supplies:

-Stool container, Small (Random), 4 oz (T288)

-Stool Collection Kit, Random (T635)

Container/Tube: Fecal container

Specimen Volume: Representative portion of feces

Collection Instructions: Collect fresh fecal specimen and submit representative sample in fecal container.

Specimen Stability Information: Refrigerated (preferred) 7 days/Frozen 7 days

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Varies	7 days

CPT Code Information:

87798

SHIGC
606221

Shigella Culture, Feces

Specimen Requirements:

Patient Preparation: The patient **should not** use antacids, barium, bismuth, antidiarrheal medication, or oily laxatives for 7 to 10 days before specimen collection.

Supplies: Culture and Sensitivity Stool Transport Vial (T058)

Container/Tube: Commercially available transport system specific for recovery of enteric pathogens from fecal specimens (15 mL of non-nutritive transport medium containing phenol red as a pH indicator, either Cary-Blair or Para-Pak C and S)

Specimen Volume: Representative portion of fecal specimen

Collection Instructions:

1. Collect fresh feces and submit 1 gram or 5 mL in container with transport medium.

2. Within 2 hours of collection, place feces in preservative.

3. Place vial in a sealed plastic bag.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

87046-Shigella Culture, Stool-with isolation and preliminary examination

87077-Bacteria Identification (if appropriate)

87153-Aerobe Ident by Sequencing (if appropriate)

87077-Additional Identification Procedure (if appropriate)

87147 x 4-Serologic Agglut Method 3 Ident (if appropriate)

87077-Ident by MALDI-TOF mass spec (if appropriate)

SQTSG 617463 Short QT Syndrome Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies.

Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for

Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to

DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81403

81406 x 2

81479

81479 (if appropriate for government payers)

SRW
82667

Short Ragweed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PSRCH
620808

Short Renal Clearance, Iohexol, Preparation

CPT Code Information:

36415 x2-Blood

81050 x3-Urine

96372-Injection

Q9961-Contrast dye

FSHRG
57542

Shrimp IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

SHRI
82677

Shrimp, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SAU
620916

Sialic Acid, Free and Total, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5mL (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 1 mL

Pediatric Volume: 0.5 mL

Collection Instructions: Collect a random urine specimen **(early morning preferred).**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Refrigerated (preferred)	90 days
	Frozen	90 days

CPT Code Information:

84275

SDEX 9180 Sickle Solubility, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD solution B), green top (heparin)

Specimen Volume: 1 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	14 days

CPT Code Information:

85660

STAT6 70554 Signal Transducer and Activator of Transcription 6 (STAT6), Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FSILS
75690

Silicon, Serum

Specimen Requirements:

*****Must submit one specimen per orders. Specimens cannot be shared between multiple orders.*****

Supplies: NMS Acid Washed "Precleaned", 10 mL (T619)

Collection Container/Tube: Royal blue top (no additive, metal-free)(serum gel/SST are **not acceptable**)

Submission Container/Tube: Trace metal-free plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions:

1. Promptly centrifuge and aliquot serum into a trace metal-free plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: Serum: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	21 days	Acid Washed Plastic (MML Supply T619)
	Ambient	14 days	Acid Washed Plastic (MML Supply T619)
	Frozen	14 days	Acid Washed Plastic (MML Supply T619)

CPT Code Information:

84285

BIR
82674

Silver Birch, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FSINS
75400

Sinemet, Serum

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Promptly centrifuge and aliquot serum into a plastic vial.

2. **Immediately freeze specimen on dry ice or results will be compromised**.

3. Send frozen.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen	6 days

CPT Code Information:

80299

SIIRO 35144 Sirolimus, Whole Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood immediately before a scheduled dose.

2. Do not centrifuge.

3. Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: Therapeutic range applies to trough specimen collected immediately prior to a.m. dose.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80195

SMAD4 SMAD4 Immunostain, Technical Component Only

613980

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SDLDL
619974

Small Dense Low Density Lipoprotein Cholesterol, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours; required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	72 hours
	Ambient	8 hours

CPT Code Information:

83722

SLL
65884

Small Lymphocytic Lymphoma, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin stained and 2 unstained for each probe set

Collection Instructions:

1. Include 1 hematoxylin and eosin-stained slide for the entire test order.

2. If individual probe sets are chosen: For each probe set ordered, 2 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

3. If a complete small lymphocytic lymphoma (SLL) panel is ordered, submit 14 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	

	Refrigerated

CPT Code Information:

88377-if 1 probe set

88377 x 2-if 2 probe sets

88377 x 3-if 3 probe sets

88377 x 4-if 4 probe sets

88377 x 5-if 5 probe sets

88377 x 6-if 6 probe sets

88377 x 7-if 7 probe sets

88377 x 8-if 8 probe sets

SM
81358

Smith (Sm) Antibodies, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235

DHCRZ
608025

Smith Lemli Opitz Syndrome, DHCR7 Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit of only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a tissue biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood**

specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

SLO 81595 Smith-Lemli-Opitz Screen, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Green top (sodium or lithium heparin)

Acceptable: Lavender top (EDTA), pearl white top (EDTA plasma gel), yellow top (ACD solution A or B)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL plasma

Collection Instructions:

1. Centrifuge and aliquot plasma into plastic vial.

2. Send plasma frozen.

Specimen Minimum Volume: Plasma: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	92 days
	Refrigerated	28 days
	Ambient	14 days

CPT Code Information:

82542

SMN1Z SMN1 Gene, Full Gene Analysis, Varies

65941

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with

1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2**.** A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 Flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81336

88233-Tissue culture, skin, or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

SMAS
609515

Smooth Muscle Antibody Screen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.8 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86015

SMAT
608956

Smooth Muscle Antibody Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a reflex. For more information see SMAS / Smooth Muscle Antibody Screen, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86015

SMOTH
70552

Smoothelin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

FCRNS 57961 Smut Corn (Ustilago maydis) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

SNAIL
82344

Snail, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SOD1Z
617740

SOD1 Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have

received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin

and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies

CPT Code Information:

81404

NAU
610734

Sodium, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: 24-hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 5 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84300

RNAUR
610785

Sodium, Random, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, 5-mL tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84300

NAS
602353

Sodium, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Ambient	14 days

CPT Code Information:

84295

SOLEF
86310

Sole, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FSFM
58015

Soluble Fibrin Monomer

Specimen Requirements:

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Platelet-poor plasma

Collection Instructions:

1. Centrifuge at 1700 x *g* for 15 minutes.

2. Aliquot 1.5 platelet-poor plasma into plastic vial making sure not to disturb the buffy coat.

3. Send frozen.

Note: **Critical frozen**. **Separate specimens must be submitted** when multiple tests are ordered.

Specimen Minimum Volume: Platelet-poor plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85366

SOLFM 602175 Soluble Fibrin Monomer, Plasma (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ALUPP / Lupus Anticoagulant Profile, Plasma

ALBLD / Bleeding Diathesis Profile, Limited, Plasma

AATHR / Thrombophilia Profile, Plasma and Whole Blood

APROL / Prolonged Clot Time Profile, Plasma

ADIC / Disseminated Intravascular Coagulation/Intravascular Coagulation and Fibrinolysis (DIC/ICF) Profile, Plasma

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85366

SLA 624092

Soluble Liver Antigen (SLA) IgG Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

SLC1Q
610061 Solute Carrier Organic Anion Transporter Family Member 1B1 (SLCO1B1) Genotype, Statin, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant or liver transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell or liver transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81328

FSOM1 Somatostatin (Somatotropin Release-Inhibiting Factor, SRIF)
75906

Specimen Requirements:

Patient Preparation:

1. Fasting: 10 to 12 hours

2. Patient **should not** be on any medications that affect insulin secretion or intestinal motility, if possible, for at least 48 hours prior to collection.

Supplies: GI Preservative (T125)

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: GI preservative tube

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL plasma

Collection Instructions:

1. Centrifuge as soon as possible and aliquot 3 mL of plasma into a plastic vial.

2. Freeze plasma immediately after separation

3. Send frozen.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	180 days
	Refrigerated	7 days

CPT Code Information:

84307

SOMAT 70553 Somatostatin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SSTR2 113597 Somatostatin Receptor 2 (SSTR2), Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However,

due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SORBU 614934 Sorbitol and Mannitol, Quantitative, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

82542

SXDBS 623688 Sorbitol and Xylitol, Quantitative, Blood Spot

Specimen Requirements:

Fasting: 8 hours, required. Infants **should not fast** overnight; collect just before next feeding.

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Card-Blood Spot Collection (Filter Paper)

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman protein Saver 903 paper, local newborn screening card, or blood collected in tubes containing EDTA (preferred) or sodium heparin and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Blood Spot	Ambient (preferred)	31 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Refrigerated	90 days	FILTER PAPER

CPT Code Information:

82542

SORD
620920

Sorbitol and Xylitol, Quantitative, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

3. Refrigerate immediately.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days

CPT Code Information:

82542

SORDB
623501

Sorbitol and Xylitol, Quantitative, Whole Blood

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium heparin)

Specimen Volume: 1 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Freeze whole blood specimens in the original tube. Frozen aliquots from well-mixed specimens are also acceptable.

3. Send frozen.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Frozen (preferred)	90 days
	Refrigerated	31 days

CPT Code Information:

82542

FSOTA
91123

Sotalol (Betapace)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days

	Ambient	72 hours

CPT Code Information:

80299

SOX10 SOX10 Immunostain, Technical Component Only
70555

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SOX11 SOX11 Immunostain, Technical Component Only
70556

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SOX9
623382

SOX9 (SRY-Box Transcription Factor) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

FSOYG 57551 Soybean IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

FSYG4 Soybean IgG4
57574

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

SOY Soybean, IgE, Serum
82886

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PBC2
620736

SP100 and GP210 Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.0 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516 x 2

SP100
620725

SP100 Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

83516

SPAGR
113392

Special Red Cell Antigen Typing, Whole Blood

Specimen Requirements:

Container/Tube:

Preferred: Pink top (EDTA)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Pediatric Volume: 3 mL Blood in pink-top or lavender-top (EDTA) tube

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Ambient	72 hours

CPT Code Information:

86905-Each red cell antigen typing (if more than one ordered)

SGUR
606565

Specific Gravity, Random, Urine

Specimen Requirements:

Container/Tube: Plastic urine container

Specimen Volume: 20 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

81005

FSPNG
57678

Spinach IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

SPIN
86312

Spinach, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

SMNCS 65574 Spinal Muscular Atrophy Carrier Screening, Deletion/Duplication Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For more information about testing patients who have received a hematopoietic stem cell, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the

cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Muscle tissue biopsy

Supplies: Muscle Biopsy Kit (T541)

Specimen Volume: 20 to 80mg

Collection Instructions: Prepare and transport specimen per instructions in Muscle Biopsy Specimen Preparation Instructions.

Specimen Stability Information: Frozen (preferred) Ambient/Refrigerated <24hours

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81329

88233 (if appropriate)

88240 (if appropriate)

SMNDX 65575 Spinal Muscular Atrophy Diagnostic Assay, Deletion/Duplication Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Direct testing of an uncultured specimen may be attempted for this test. Contact the laboratory at 800-533-1710 if direct testing is desired.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81329

88235 (if appropriate)

88240 (if appropriate)

88233 (if appropriate)

88240 (if appropriate)

81265 (if appropriate)

SBULB 35542 Spinobulbar Muscular Atrophy (Kennedy Disease), Molecular Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone

marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic,

Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81204-*AR (*androgen receptor)(eg, spinal and bulba muscular atrophy, Kennedy disease, X chromosome inactivation) gene analysis; characterization of alleles (eg, expanded size or methylation status)

SCAP 609505 Spinocerebellar Ataxia Repeat Expansion Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4

days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81178

81179

81180

81181

81184

81479 (if appropriate for government payers)

SCARA 609698 Spinocerebellar Ataxia Type 1, 2, 3, 6, or 7, Repeat Expansion Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal

testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional information:

1. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid.

2. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Refrigerated

Additional Information:

1. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

2. All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Acceptable:

Specimen Type: Confluent cultured cells

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information: All prenatal specimens must be accompanied by a maternal blood specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: Amniotic fluid: 10 mL
Blood: 0.5 mL
Chorionic villi: 5 mg

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

88233-Fibroblast Culture (if appropriate)

88235-Amniotic Fluid Culture (if appropriate)

88240-Cryopreservation (if appropriate)

81265-Maternal Cell Contamination (if appropriate)

81178 (if appropriate)

81179 (if appropriate)

81180 (if appropriate)

81181 (if appropriate)

81184 (if appropriate)

SFGP 83679 Spotted Fever Group Antibody, IgG and IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86757 x 2

SPRU
82394

Spruce, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SCCA
610033

Squamous Cell Carcinoma Antigen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into a plastic vial. **Do not submit in original tube**.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Frozen (preferred)	90 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

86316

SQUA
82797

Squash, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SQUID
82631

Squid, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SSAB
82403

SS-A and SS-B Antibodies, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235 x 2

SSA
81360

SS-A/Ro Antibodies, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235

SSB
81359

SS-B/La Antibodies, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235

STLPC 83916 St. Louis Encephalitis Antibody Panel, IgG and IgM, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see ABOPC / Arbovirus Antibody Panel, IgG and IgM, Spinal Fluid.

Container/Tube: Sterile vial

Preferred: Vial number 2

Acceptable: Any vial

Specimen Volume: 0.7 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86653 x 2

FSTAB
57891

Stachybotrys chartarum/atra IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

STRC
83316

Standard Renal Clearance, Plasma and Random Urine

Specimen Requirements:

Contact MLI for approval prior to ordering this test.

Both plasma and urine are required.

Specimen Type: Plasma

Container/Tube: 6-mL Green top (sodium heparin)

Specimen Volume: 3-4 mL

Collection Instructions: Label specimen as plasma.

Specimen Type: Urine

Container/Tube: Sarstedt 5 mL Aliquot Tube (T914)

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine collection.

2. No preservative.

3. Label specimen as urine.

Specimen Minimum Volume: 2 mL plasma/5 mL urine

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Heparin	Refrigerated (preferred)	7 days
	Frozen	7 days
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

82542 x 7

STEM
82696

Stemphyllium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

STER
82079

Sterols, Plasma

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Green top (sodium or lithium heparin)

Acceptable: Lavender top (EDTA), pearl white top (EDTA plasma gel), yellow top (ACD solution A or B)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL plasma

Collection Instructions:

1. Centrifuge specimen and aliquot plasma into plastic vial.

2. Send plasma frozen.

Specimen Minimum Volume: Plasma: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	92 days
	Refrigerated	28 days
	Ambient	14 days

CPT Code Information:

82542

STFR1 sTfR (Soluble Transferrin Receptor), Serum
623578

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For red-top tubes, aliquot serum into a plastic vial prior to sending.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	7 days

CPT Code Information:

84238

SPPS Stiff-Person Spectrum Disorders Evaluation, including Progressive Encephalomyelitis with Rigidity and Myoclonus, Serum
614599

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 2

86341

0431U

84182 (if appropriate)

86256 (if appropriate)

86256 (if appropriate)

SPPC
614600

Stiff-Person Spectrum Disorders Evaluation, including Progressive Encephalomyelitis with Rigidity and Myoclonus, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 4 mL

Specimen Minimum Volume:

2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x 2

86341

0431U

84182 (if appropriate)
86256 (if appropriate)

86256 (if appropriate)

SPPCI
614602

Stiff-Person Spectrum Disorders including Progressive Encephalomyelitis with Rigidity and Myoclonus, Interpretation, Spinal Fluid

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

SPPSI
614601

Stiff-Person Spectrum Disorders, including Progressive Encephalomyelitis with Rigidity and Myoclonus, Interpretation, Serum

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

INSEC
31765

Stinging Insects Allergen Profile, Serum

Specimen Requirements:

Supplies:

Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003 x 5

FSTBG 57656 Strawberry IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

STBY
82676

Strawberry, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SABP
86537

Streptococcal Antibodies Profile, Serum

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, preferred but not required

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

86060

86215

GAPCR 610254 Streptococcus Group A, Molecular Detection, PCR, Throat

Specimen Requirements:

Patient Preparation: Do not collect throat swab specimens immediately after use of antiseptic mouthwash.

Specimen Type: Throat swab

Supplies: BD E-Swab (T853)

Container/Tube:

Preferred: BD Liquid Amies Elution Swab (ESwab)

Acceptable: Copan Liquid Amies Elution Swab (ESwab)

Specimen Volume: Entire collection/1 ESwab

Collection Instructions:

1. **Must be collected and transported using the ESwab collection kit.**

2. Using a tongue blade to hold the tongue down, firmly rub the swab over the tonsils, posterior pharynx, and any areas of inflammation with exudate. Ensure that areas with exudate or inflammation are swabbed.

3. Avoid the tongue, teeth, and gums as the swab is withdrawn.

4. Insert the swab back into the ESwab transport tube and break the swab at the indicated score line.

5. Label specimen according to policy.

6. Send refrigerated. **Do not freeze**.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	6 days
	Ambient	48 hours

CPT Code Information:

87651

SPNEU Streptococcus pneumoniae Antigen, Random, Urine

83150

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 2 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

3. Excessively bloody or very turbid specimens containing protein, cells, or particulates will be canceled, as they can inhibit the function of the test.

4. **Do not centrifuge to remove particulates**.

5. Specimens with any dyes or unnatural color are not acceptable and will be canceled.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	24 hours

CPT Code Information:

87899

SPNC
89971

Streptococcus pneumoniae Antigen, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Submit specimen collected in vial 2, if possible. If not possible, note the vial from which the aliquot was obtained.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

87899

PN23M
619921

Streptococcus pneumoniae IgG Antibodies, 23 Serotypes, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86581

PNTO
608969

Streptococcus pneumoniae IgG Antibodies, Total, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86317

PNTOR 608970 Streptococcus pneumoniae IgG Antibodies, Total, with Reflex, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86317

86317 x 23 (if appropriate)

MSTC 80749 Strict Criteria Sperm Morphology for Infertility Diagnosis and Treatment, Semen

Specimen Requirements:

Container/Tube: Semen Analysis Kit (T178)

Collection Instructions: Patient should have 2 to 7 days of sexual abstinence at the time of semen collection for accurate results.

Specimen Volume: Total ejaculate

Specimen Minimum Volume: A minimum count is needed. Lab will determine.

Transport Temperature:

Specimen Type	Temperature	Time
Semen	Ambient	

CPT Code Information:

89398

MSTC1 35184 Strict Criteria Sperm Morphology for Infertility Diagnosis and Treatment, Semen

Specimen Requirements:

Patient Preparation: Patient should have 2 to 7 days of sexual abstinence at the time of semen collection for accurate results.

Container/Tube: Slides

Specimen Volume: 2 slides-10 microL of liquefied semen on each slide

Collection Instructions:

1. If sperm concentration is <10 million/mL, centrifuge the specimen at 300 x G for 10 minutes before making slides.

2 Label 2 frosted slides **in pencil** with the patient's first and last name and the date of specimen collection. No adhesive labels.

3. Allow the semen to liquefy for 30 minutes.

4. Place 10 mcL of liquefied semen on the label end of each slide, and evenly smear the specimen using a plain slide (this process is the same as making a blood smear).

5. Allow the smears to air dry for 15 minutes before placing both slides into 1 slide holder for shipment.

Specimen Minimum Volume: A minimum count of sperm is needed; lab will determine

Transport Temperature:

Specimen Type	Temperature	Time
Semen	Ambient	

CPT Code Information:

89398

STRNG
63866

Strongyloides Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86682

FSTYR
91094

Styrene, Occupational Exposure, Blood

Specimen Requirements:

Collect 2 tubes green-top (sodium heparin) whole blood. Send 20 mL sodium heparin whole blood refrigerated. Collect specimen at end of shift or prior to next shift. Tubes should be filled to prevent loss of volatile compound into headspace.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
WB Sodium Heparin	Refrigerated (preferred)	14 days
	Frozen	180 days

CPT Code Information:

84600

SUAC 83635 Succinylacetone, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood Spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tube containing EDTA and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. At least 1 spot should be complete (ie, unpunched).

3. Let blood dry on the Blood Spot Collection Card (T493) at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Ambient (preferred) 7 days/Refrigerated 14 days/Frozen 90 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Acceptable

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate 6 days

Specimen Minimum Volume: Blood spots: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

84510

82542

82542 (if appropriate for government payers)

SUDC
606930

Sudden Cardiac Death Pathology Consultation

Specimen Requirements:

1. The heart specimen is required for testing.

2. Send whole blood in addition to heart specimen, if available.

Specimen Type: Heart specimen

Supplies:

-Pathology Packaging Kit (T554)

-Ambient Shipping Box, 5 lb (T970)

Specimen Volume: Entire specimen

Collection Instructions:

1. Fix entire specimen in formalin for a minimum of 24 hours.

2. Following fixation, lightly wrap specimen in formalin-soaked paper towel, place in a leak-proof sealable bag with a small amount of formalin, seal, and ship ambient in a sturdy shipping container.

Additional Information: Paraffin block may be accepted.

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen.

Specimen Minimum Volume: Heart specimen: See Specimen Required; Blood: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
SUDC Study Specimen	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

88037

FSCNE 57543 Sugar Cane (Saccharum officinarum) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

SBSE
82382

Sugarbeet Seed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SFZ
8238

Sulfamethoxazole, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (gel tubes/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum for a peak level should be collected 60 minutes after dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

SULFU
606479

Sulfate, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Container/Tube: Plastic, 5-mL urine tube

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Collection instructions with **no preservative:**

-Specimen must be kept refrigerated during and after collection.

3. Collection instructions **with preservative:**

a. If preservative is **not** added **before** collection, the specimen **must be kept refrigerated until preservative is added**.

b. Preservative must be added **within 4 hours** of the end of collection.

4. Specimen pH should be between 4.5 and 8 and will stay in this range if kept refrigerated. Specimens with pH >8 indicate bacterial contamination, and testing will be cancelled. Do not attempt to adjust pH as it will adversely affect results.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

84392

FSLFU 57710 Sulfonylurea Screen, Urine

Specimen Requirements:

Collection Container: Plastic urine container

Specimen Volume: 5 mL

Collection Instructions:

1. Collect 5 mL random urine without preservative.
2. Ship refrigerated in a plastic container.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80377

FSUNG
57681

Sunflower Seed IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

SUNFS
82813

Sunflower Seed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SUNF
82615

Sunflower, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SUP24 Supersaturation Profile, 24 Hour, Urine

616180

Specimen Requirements:

Patient Preparation:

X-ray dyes and contrast media will affect uric acid test results.

-If a kidney X-ray with dye or computerized tomography (CT) scan with contrast has been performed, patient should wait a minimum of 1 day before starting collection.

-If a cholangiography (bile duct X-ray) has performed, patient should wait 7 days before starting collection.

-Urine must be collected before tablets have been taken for gallbladder X-ray, otherwise patient should wait 7 days before starting collection.

Supplies: Diazolidinyl Urea (Germall) 5.0 mL (T822)

Collection Container/Tube: 24-hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 35 mL

Collection Instructions:

1. Add 5 mL of diazolidinyl urea as preservative at start of collection or refrigerate specimen during and after collection.

2. Collect urine for 24 hours.

3. Specimen pH should be between 4.5 and 8 and will stay in this range if diazolidinyl urea is added at the beginning of the collection or kept refrigerated during the entirety of the collection. Specimens with pH above 8 indicate bacterial contamination, and testing will be canceled. Do not attempt to adjust pH as it will adversely affect results.

4. **If not using Germall, the specimen must be kept refrigerated** during the entirety of the collection and sent frozen.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 16 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

82340-Calcium

82436-Chloride

82507-Citrate excretion

82570-Creatinine

83735-Magnesium

83935-Osmolality

83945-Oxalate

83986-pH

84105-Phosphorus

84133-Potassium

84300-Sodium

84392-Sulfate

84560-Uric acid

82140-Ammonium

84540-Urea Nitrogen

SUPRA 616375 Supersaturation Profile, Random, Urine

Specimen Requirements:

Supplies:

-Urine Tubes, 10 mL (T068)

-Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: 90 mL urine container

Submission Container/Tube: 2 x 10 mL plastic urine tubes and 5 x 5 mL plastic urine tubes

Specimen Volume: 40 mL

Collection Instructions:

1. Collect a random urine specimen, mix collection container thoroughly and divide the urine into 7 plastic tubes (2 x 10 mL and 5 x 5 mL).

2. Refrigerate specimen after collection. Specimen pH should be between 4.5 and 8 and will stay in this range if kept refrigerated. Specimens with pH above 8 indicate bacterial contamination, and testing will be canceled. **Do not** attempt to adjust pH as it will adversely affect results.

Specimen Minimum Volume: 16 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

82310-Calcium

82436-Chloride

82507-Citrate excretion

82570-Creatinine

83735-Magnesium

83935-Osmolality

83945-Oxalate

83986-pH

84105-Phosphorus

84133-Potassium

84300-Sodium

84392-Sulfate

84560-Uric acid

82140-Ammonium

SSH24 Supersaturation Profile, Self-Collect, 24 Hour, Urine

616228

Specimen Requirements:

Patient Preparation:

X-ray dyes and contrast media will affect uric acid test results.

-If a kidney X-ray with dye or computerized tomography (CT) scan with contrast has been performed, the patient should wait at least 24 hours before starting collection.

-If a cholangiography (bile duct X-ray) has been performed, the patient should wait 7 days before starting collection.

-If patient has taken tablets for a gallbladder X-ray, the patient should wait 7 days before starting collection.

Supplies: SSH24 Home Collection Kit (T921)

Collection Container/Tube: 24-hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 35 mL

Collection Instructions:

1. Order test each time the patient is to collect a 24-hour urine specimen at home and mail the specimen directly to Mayo Clinic Laboratories.

2. Order should be placed a minimum of 3 days prior to desired date of collection.

3. Enter patient's address information for each order created, including street address, city, state abbreviation, zip code, country, and home phone number.

4. For each order, the SSH24 Home Collection kit will be mailed directly to the patient for self-collection. This kit contains instructions for how to properly collect and transport the specimen after collection.

Specimen Minimum Volume: 16 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

82340-Calcium

82436-Chloride

82507-Citrate excretion

82570-Creatinine

83735-Magnesium

83935-Osmolality

83945-Oxalate

83986-pH

84105-Phosphorus

84133-Potassium

84300-Sodium

84392-Sulfate

84560-Uric acid

82140-Ammonium

84540-Urea Nitrogen

SNS
82594

Supplemental Newborn Screen, Blood Spot

Specimen Requirements:

Patient must be older than 24 hours and less than 1 week of age.

Supplies: Card-Blood Spot Collection Filter Paper (T493)

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, Munktell, PerkinElmer 226 filter paper, or local newborn screening card

Specimen Volume: 3 Blood spots

Collection Instructions:

1. Do not use device or capillary tube containing EDTA to collect specimen.

2. Completely fill at least 3 circles on the filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on the Blood Spot Collection Card at ambient temperature in a horizontal position for 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)		FILTER PAPER
	Frozen		FILTER PAPER
	Refrigerated		FILTER PAPER

CPT Code Information:

83789

TB2LN 34550 Susceptibility, Mycobacterium tuberculosis Complex, Second Line, Varies

Specimen Requirements:

Specimen Type: Organism

Supplies: Infectious Container, Large (T146)

Container/Tube: Middlebrook 7H10 agar slant

Specimen Volume: Isolate

Collection Instructions: Organism must be in pure culture, actively growing.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87186-Susceptibility, Mtb Cx, 2nd Line

SV40
618827

SV40 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SGUM
82483

Sweet Gum, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SPOT
82799

Sweet Potato, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

VERG
82909

Sweet Vernal Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

HBRM
622927

SWI/SNF Chromatin Remodeling Complex Subunit ATPase 2 (SMARCA2 or BRM) Immunostain, Technical Component

Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

SWORD
82346

Swordfish, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

SYNAP
70557

Synaptophysin (SYNAPTO) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

SS18F 35303 Synovial Sarcoma (SS), 18q11.2 (SS18 or SYT) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

SSX18 Synovial Sarcoma Panel, Technical Component Only

610609

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FSCBB 75967 Synthetic Cannabinoid Metabolites Confirmation (Qualitative)-Expanded, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see FSCBA / Synthetic Cannabinoid Metabolites Screen-Expanded, Urine

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	30 days

CPT Code Information:

80352

FSCBA 75966 Synthetic Cannabinoid Metabolites Screen-Expanded, Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Container/Tube: Plastic, preservative-free urine container

Specimen Volume: 3 mL

Collection Instructions:

1. Collect 3 mL random urine specimen without preservative.

2. Send refrigerated in a plastic preservative-free urine container.

Specimen Minimum Volume: 1.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	30 days

CPT Code Information:

80307

SGSU
81035

Synthetic Glucocorticoid Screen, Random, Urine

Specimen Requirements:

Supplies: Urine tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	14 days

CPT Code Information:

80299

SGSS
81031

Synthetic Glucocorticoid Screen, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Refrigerated	7 days
	Ambient	24 hours

CPT Code Information:

80299

TPPA 61480 Syphilis Antibody, Treponema pallidum Particle Agglutination, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86780

NSYPH 616862 Syphilis IgG Enzyme Immunoassay, Serum

Specimen Requirements:

Collection Container/Tube: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86780

SYPH1 616860 Syphilis IgG with Reflex, Enzyme Immunoassay, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86780-Syphilis IgG Screen

86592-Syphilis Rapid Plasma Reagin Screen (if appropriate)

86593-Rapid Plasma Reagin Titer (if appropriate)

86780-Syphilis Antibody by TP-PA (if appropriate)

SYPHT
603264

Syphilis Total Antibody with Reflex, Serum

Specimen Requirements:

Collection Container/Tube: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86780

86780-Syphilis Antibody by TP-PA (if appropriate)

86593-Rapid Plasma Reagin Titer (if appropriate)

SSCP
622520

Systemic Sclerosis Criteria Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86039

86235

83516

TBNY
82589

T, B and NK Lymphocyte Quantitation, New York, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimen collection be performed at the same time of day.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	52 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86355

86357

86359

86360

TBET
70559

T-Box Expressed in T Cells (TBET) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

COGTF 113529 T-Cell Acute Lymphoblastic Leukemia/Lymphoma (ALL), Children's Oncology Group Enrollment Testing, FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x2, 88275 x1, 88291 x1- FISH Probe, Analysis, Interpretation; 1 probe set

88271 x2, 88275x1 - FISH Probe, Analysis; each additional probe set (if appropriate)

TALAF 609557 T-Cell Acute Lymphoblastic Leukemia/Lymphoma (ALL), FISH, Adult, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x20, 88275x10, 88291x1-FISH Probe, Analysis, Interpretation; 10 probe sets

88271x2, 88275x1-FISH Probe, Analysis; each additional probe set (if appropriate)

TALMF 614323 T-Cell Acute Lymphoblastic Leukemia/Lymphoma (ALL), Specified FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88275x1, 88291x1-FISH Probe, Analysis, Interpretation; 1 probe set

88271x2, 88275x1 - FISH Probe, Analysis; each additional probe set (if appropriate)

BTIA1
605160

T-Cell Intracellular Antigen 1 (TIA-1) Immunostain, Bone Marrow, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TIA1
70566

T-Cell Intracellular Antigen 1 (TIA-1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TCL1A
70561

T-Cell Leukemia/Lymphoma Protein 1A (TCL1A) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However,

due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

TLBLF 65413 T-Cell Lymphoblastic Leukemia/Lymphoma, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin stained and 2 unstained for each probe set

Collection Instructions:

1. Include 1 hematoxylin and eosin-stained slide for the entire test order.

2. If individual probe sets are chosen: For each probe set ordered, submit 2 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

3. If a complete T-lymphoblastic leukemia/lymphoma (TLBL) panel is ordered: submit 16 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88291-DNA probe, each (first probe set), Interpretation and report

88271 x 2-DNA probe, each; each additional probe set (if appropriate)

88271-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271 x 2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271 x 3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

TLPFD 622430 T-Cell Lymphoma Panel, Diagnostic, FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2-3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x8, 88275 x4, 88291 x1- FISH Probe, Analysis, Interpretation; 4 probe set

88271 x2, 88275 x1 - FISH Probe, Analysis; each additional probe set (if appropriate)

TLYM 65911 T-Cell Lymphoma, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 2 unstained for each probe set

Collection Instructions:

1. Include 1 hematoxylin and eosin-stained slide for the entire test order.

2. For each probe set ordered, 2 consecutive, unstained, 5 micron-thick sections placed on positively charged slides.

3. If ordering TCL1A or TRA, 4 unstained slides are necessary; the break-apart TCL1A and TRA break-apart probe sets are performed simultaneously.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88377 (if 1 probe set)

88377 x 2 (if 2 probe sets)

88377 x 3 (if 3 probe sets)

88377 x 4 (if 4 probe sets)

88377 x 5 (if 5 probe sets)

88377 x 6 (if 6 probe sets)

88377 x 7 (if 7 probe sets)

88377 x 8 (if 8 probe sets)

TLPMF
614346

T-Cell Lymphoma, Specified FISH, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 2 to 3 mL

Collection Instructions:

1. It is preferable to send the first aspirate from the bone marrow collection.

2. Invert several times to mix bone marrow.

3. Send bone marrow in original tube. **Do not aliquot.**

Acceptable

Specimen Type: Whole Blood

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (sodium heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Bone marrow: 1 mL; Whole blood: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88275 x1, 88291x1- FISH Probe, Analysis, Interpretation; 1 probe set

88271x2, 88275 x1 - FISH Probe, Analysis; each additional probe set (if appropriate)

TCRF1 70560 T-Cell Receptor Beta (TCR Beta F1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TRBC3 622845 T-Cell Receptor Constant Beta Chain (TRBC1) +CD3 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344-TC

TRBC1 622852 T-Cell Receptor Constant Beta Chain (TRBC1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

TCRGD 70562 T-Cell Receptor Delta Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TRECS 610068 T-Cell Receptor Excision Circles Analysis, Blood

Specimen Requirements:

For serial monitoring, it is recommended to perform specimen collection at the same time of day, if possible.

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Container/Tube: Lavender top (EDTA)

Specimen Volume:

Adults: 10 mL

Pediatrics

-Preferred volume for >1 year: 5 mL

-Preferred volume for < or =1 year old: 3 mL

Collection Instructions:

1. Do not collect specimen using a butterfly needle.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Adults: 10 mL
Pediatrics: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	48 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81479-Unlisted molecular pathology procedure

TCGR
83122

T-Cell Receptor Gene Rearrangement, PCR, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81340-TCB (T cell antigen receptor, beta) (eg, leukemia and lymphoma), gene rearrangement analysis to detect abnormal clonal population(s); using amplification methodology (eg, PCR)

81342-TCG (T cell receptor, gamma) (eg, leukemia and lymphoma), gene rearrangement analysis, evaluation to detect abnormal clonal population(s)

TCGBM T-Cell Receptor Gene Rearrangement, PCR, Bone Marrow

31139

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81340-TCB (T cell antigen receptor, beta) (eg, leukemia and lymphoma), gene rearrangement analysis to detect abnormal clonal population(s); using amplification methodology (eg, PCR)

81342-TCG (T cell receptor, gamma) (eg, leukemia and lymphoma), gene rearrangement analysis, evaluation to detect abnormal clonal population(s)

TCGET
802124

T-Cell Receptor Gene Rearrangement, PCR, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Tissue slides

Slides: 4 unstained

Collection Instructions: Submit 4 consecutive, positively charged, unbaked, unstained slides with 10-micron thick sections

Acceptable

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block containing neoplastic cells

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue, Paraffin	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81340-TCB (T cell antigen receptor, beta) (eg, leukemia and lymphoma), gene rearrangement analysis to detect abnormal clonal population(s); using amplification methodology (eg, PCR)

81342-TCG (T cell receptor, gamma) (eg, leukemia and lymphoma), gene rearrangement analysis, evaluation to detect abnormal clonal population(s)

TCGRV
31140

T-Cell Receptor Gene Rearrangement, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Paraffin-embedded tissue

Container/ Tube: Paraffin block

Collection Instructions:

1. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

2. Indicate specimen source.

3. Include pathology report.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 Unstained slides

Container/Tube: Transport in plastic slide holders

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

3. Indicate specimen source.

4. Include pathology report.

Specimen Stability Information: Ambient

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Body fluid

Sources: Pleural, peritoneal, vitreous and spinal fluid

Container/Tube: Sterile container

Specimen Volume: At least 5 mL

Collection Instructions:

1. If the volume is large, pellet cells prior to sending.

2. Send less volume at ambient temperature or as a frozen cell pellet.

3. Specify the type of fluid being submitted.

Specimen Stability Information:

Body fluid: Ambient 4 days/Refrigerated/Frozen

Cell pellet: Frozen

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions:

1. Freeze tissue within 1 hour of collection.

2. Indicate specimen source.

Specimen Stability Information: Frozen

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of DNA on label.

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods

are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Body fluid: 1 mL; Frozen tissue: 50 mg; Extracted DNA: 50 microliters (mcL) at 20 ng/mcL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81340-TCB (T cell antigen receptor, beta) (eg, leukemia and lymphoma), gene rearrangement analysis to detect abnormal clonal population(s), using amplification methodology (eg, PCR)

81342-TCG (T cell receptor, gamma) (eg, leukemia and lymphoma), gene rearrangement analysis, evaluation to detect abnormal clonal population(s)

TCRB 610067 T-Cell Receptor V-Beta Repertoire Analysis, Spectratyping, Blood

Specimen Requirements:

For serial monitoring, it is recommended to perform specimen collection at the same time of day, if possible.

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Specimen Type: Blood

Container/Tube: Lavender top (EDTA)

Specimen Volume:

Adults: 10 mL

Pediatrics:

-Preferred volume for >1 year: 3 mL

-Preferred volume for < or =1 year: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: Adults: 5 mL

Pediatrics: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	48 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81340-TRG (T cell antigen receptor, beta) (eg, leukemia and lymphoma), gene rearrangement analysis to detect abnormal clonal population(s); using amplification methodology (eg, polymerase chain reaction)

TCP
89319

T-Cell Subsets, Naive, Memory, and Activated, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86356 x 7

TREGS
89318

T-Cell Subsets, Regulatory (Tregs), Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86359

86361

CYTH1 T-Helper Cell Type 1 Cytokine Panel, Plasma

622036

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Plasma

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice.

2. Within 2 hours of collection, centrifuge at 4 degrees C, 1500 x *g* for 10 minutes and aliquot plasma into a plastic vial

3. Freeze immediately.

Additional Information:

1. Plasma for this test cannot be shared with other tests. Submit specimen for this test in its own vial.

2. This test cannot be added on to a previously collected specimen.

Specimen Minimum Volume: Plasma: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	21 days

CPT Code Information:

83520 x5

CYTH2 T-Helper Cell Type 2 Cytokine Panel, Plasma
622929

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice.

2. Within 2 hours of collection, centrifuge at 4 degrees C, 1500 x *g* for 10 minutes and aliquot plasma into a plastic vial.

3. Freeze immediately.

Additional Information:

1. **Plasma for this test cannot be shared with other tests**. Submit specimen for this test in its own vial.

2. **This test cannot be added onto a previously collected specimen**.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	21 days

CPT Code Information:

83520 x4

TPIT T-PIT Immunostain, Technical Component Only
607889

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

T3FR
621321

T3 (Triiodothyronine), Free, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days

CPT Code Information:

84481

RT3
9405

T3 (Triiodothyronine), Reverse, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.8 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

84482

T3
8613

T3 (Triiodothyronine), Total, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

84480

FRT3S
75752

T3 Reverse, LC/MS/MS

Specimen Requirements:

Collection Container/Tube: Red top or serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Allow blood to clot at room temperature for 30 minutes.

2. Within 24 hours of collection, centrifuge and aliquot serum immediately into a plastic vial.

3. Send frozen.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

84482

FRT4D
8859

T4 (Thyroxine), Free, Dialysis, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2.6 mL Serum

Collection Instructions:

1. Collect specimen immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	21 days
	Ambient	7 days

CPT Code Information:

84439

FRT4
8725

T4 (Thyroxine), Free, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection do not take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

84439

T4FT4
36108

T4 (Thyroxine), Total and Free, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.625 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

84436

84439

T4
8724

T4 (Thyroxine), Total Only, Serum

Specimen Requirements:

Patient Preparation: For 4 to 6 weeks before specimen collection, patient should not be taking lipid-lowering medications containing dextrothyroxine.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

84436

FFT4F
91548

T4, Free, Direct Dialysis

Specimen Requirements:

Patient Preparation:

Fasting: 8-16 hours, preferred but not required

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collections Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	35 days
	Ambient	4 days

CPT Code Information:

84439

TAKRO Tacrolimus, Blood
35145

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood immediately before a scheduled dose.

2. Do not centrifuge.

3. Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: Therapeutic range applies to trough specimens collected

immediately prior to a.m. dose.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

80197

TACPK 88157 Tacrolimus, Peak, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Do not centrifuge.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

80197

TAPEN 62594 Tapentadol and Metabolite, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

G0480

80372 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

TAPNS 624021 Tapentadol and Metabolite, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.0 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80372 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FIOCA
57944

Tapioca IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days

	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

TBSU
604262

Targeted Benzodiazepine Screen, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

G0480

80347 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

TOSU
604261

Targeted Opioid Screen, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

G0481
80364 (if appropriate for select payers)

TSPU
610272

Targeted Stimulant Screen, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Refrigerated (preferred)	14 days
	Frozen	28 days

CPT Code Information:

G0480

80326 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

TARR
82486

Tarragon, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TRAP
70570

Tartrate-Resistant Acid Phosphatase (TRAP) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TAU3
70628

TAU 3 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TAU4
70627

TAU 4 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TAUI
70558

TAU Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HEXAN
622610

Tay-Sachs Disease, HEXA Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic

donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

FGTEA
57684

Tea IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red Top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

TEA
82625

Tea, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TELDP
619886

Telomere Biology Disorders Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing, Chorionic Villi/Products of Conception/Tissue. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. **Cultured cells from a prenatal specimen will not be accepted**.

Specimen Stability Information: Ambient (preferred)/Refrigerated (<24 hours)

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing, Chorionic Villi/Products of Conception/Tissue. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Minimum Volume: Whole blood: 1 mL; Skin biopsy or cultured fibroblasts: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233- Tissue culture, skin, solid tissue biopsy (if appropriate)

88240- Cryopreservation (if appropriate)

TEMZS
623017

Temazepam and Oxazepam, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

TDT
70563

Terminal Deoxynucleotidyl Transferase (TdT) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

TERTD
618208

TERT Promoter Mutation Analysis, Droplet Digital PCR, Tumor

Specimen Requirements:

This assay requires at least 5% nuclei of tumor cells/cells of interest.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue144 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent nuclei of tumor cells/cells of interest.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-Cytology fixatives: Cytology smears fixed in alcohol and thin preps fixed with CytoLyt.

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81345

88381-Microdissection, manual

FFTFT 75751 Testosterone, Free (Dialysis) and Total, MS

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 1.8 mL

Collection Instructions:

1. Allow blood to clot at room temperature. Centrifuge and immediately aliquot 1.8 mL of serum into a plastic vial.

2. Send refrigerate

Specimen Stability Information: Refrigerated (preferred) 21 days/Ambient 7 days/Frozen 2 years

Specimen Minimum Volume: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	21 days
	Ambient	7 days
	Frozen	

CPT Code Information:

84402

84403

FFTST 75642 Testosterone, Free and Weakly Bound, With Total Testosterone, LC/MS-MS

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions:

1. Within one hour of collection, centrifuge and aliquot 2 mL serum into a plastic vial.

2. Ship refrigerated.

Specimen Minimum Volume: 1.4 mL (does not allow for repeat testing)

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

84410

FFBTT
75934

Testosterone, Free, Bioavailable, and Total, MS

Specimen Requirements:

Caution: Administration of STRENSIQ may interfere in certain assays and may falsely elevate values. For patients receiving STRENSIQ, consideration should be given to using alternate methods.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 2.8 mL Serum

Collections Instructions:

1. Allow blood to clot at room temperature.

2. Centrifuge and aliquot 2.8 mL of serum into a plastic vial.

3. Send refrigerate.

Specimen Minimum Volume: Serum: 1.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

82040

84270

84403

TTBS
80065

Testosterone, Total and Bioavailable, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	60 days

CPT Code Information:

84403

84410

TGRP
8508

Testosterone, Total and Free, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	60 days

CPT Code Information:

84402

84403

TTFB
83686

Testosterone, Total, Bioavailable, and Free, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 3.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	60 days

CPT Code Information:

84402

84403

84410

TTST
8533

Testosterone, Total, Mass Spectrometry, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.215 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	60 days

CPT Code Information:

84403

TTIGS
36667

Tetanus Toxoid IgG Antibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86317

TGOGF 621841 TGFBR3 (1p22), OGA (10q24) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tumor tissue block. Blocks prepared with alternative fixation methods may be acceptable; provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Bone specimens that have been decalcified will be attempted for testing, but the success rate is approximately 50%.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin stained and 6 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 6 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291 DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

THEV1 608085 Thalassemia and Hemoglobinopathy Evaluation, Blood and Serum

Specimen Requirements:

Blood and serum are required.

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 15 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Type: Serum

Patient Preparation: For 12 hour before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions:

1. Within 2 hours of collection, serum gel tubes should be centrifuged.

2. Within 2 hours of collection, red-top tubes should be centrifuged and the serum aliquoted into a plastic vial.

3. Label specimen as serum.

Specimen Minimum Volume: Blood: 2.5 mL
Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated	7 days
Whole Blood EDTA	Refrigerated	7 days

CPT Code Information:

83020-26-Hemoglobinopathy Interpretation

83020-Hb Variant, A2 and F Quantitation

83021

82728

82664 (if appropriate)

83068 (if appropriate)

83789 (if appropriate)

88184 (if appropriate)

THEV0 608092 Thalassemia Summary Interpretation, Blood (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	

TLU 8603 Thallium, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to

potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium or iodine containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 10 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. Aliquot 10 mL into a plastic 10-mL urine tube (T068) or a clean, plastic aliquot container with no metal cap or glued insert.

4. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: For multiple collections, see Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83018

TLB
8149

Thallium, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based

metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube:

Preferred: Royal blue-top BD Vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 1 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83018

TLUCR 615255 Thallium/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to interfere with most metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap

or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 7 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.3mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

83018

82570

FHCCF
75831

THC Confirmation, MS, SP

Specimen Requirements:

Submit only 1 of the following specimens:

Serum:

Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable**. Spin down and send 5mL of serum refrigerated in plastic vial.

Plasma:

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable**. Spin down and send 5 mL sodium heparin plasma refrigerated in plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days

CPT Code Information:

80349

G0480 (if appropriate)

FMARX THC, MS, WB/SP Rfx (Non-Orderable)
75832

Specimen Requirements:

Only orderable as a reflex from FMARI

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

THEO Theophylline, Serum
8661

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For serum gel tubes, serum may sit on gel refrigerated or aliquoted into a plastic vial.

3. For red top tubes, aliquot serum immediately into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80198

TDP 42356 Thiamine (Vitamin B1), Whole Blood

Specimen Requirements:

Patient Preparation:

1. Fasting: 12 hours, required; Infants should have specimen collected before next feeding, water can be taken as needed

2. For 12 hours before specimen collection, patient **should not** take vitamin supplements.

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Amber vial

Specimen Volume: 4 mL

Collection Instructions:

1. Invert 8 to 10 times to mix whole blood.

2. Transfer whole blood into amber vial or tube and freeze within 24 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Frozen	28 days	LIGHT PROTECTED

CPT Code Information:

84425

DTHPV 70336 ThinPrep Diagnostic with Human Papillomavirus (HPV) Reflex, Varies

Specimen Requirements:

Patient Preparation: For optimal interpretation, Papanicolaou smears should be collected near the middle of the menstrual cycle. No douching, lubricant use, and sexual intercourse for 24 hours prior to specimen collection.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap Test, regardless of the volume of the aliquot (maximum aliquot volume=4 mL).

Submit only 1 of the following specimens:

Specimen Type: Cervical

Supplies: Thin Prep Media with Broom Kit (T056)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm water to warm and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

7. Place labels on the vial and on the bag.

Specimen Type: Ectocervix and endocervix

Supplies: Thin Prep Media with Spatula and Brush Kit (T434)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm water to warm and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire exocervix while maintaining tight contact with exocervical surface.

2. Rinse spatulas quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate one-quarter or one-half turn in 1 direction. **Do not over rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

9. Place labels on the vial and on the bag.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	THIN PREP
	Refrigerated	42 days	THIN PREP

CPT Code Information:

88142

88141(if appropriate)

TPRPD 70333 ThinPrep Diagnostic, Varies

Specimen Requirements:

Patient Preparation: For optimal interpretation, Pap smears should be collected near the middle of the menstrual cycle. Avoid douching, lubricant use, or sexual intercourse for 24 hours prior to specimen collection.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap test, regardless of the volume of the aliquot (maximum aliquot volume: 4 mL).

Submit only 1 of the following specimens:

Specimen Type: Cervical

Supplies: Thin Prep Media with Broom Kit (T056)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm water to warm and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

7. Place labels on the vial and on the bag.

Specimen Type: Ectocervix and endocervix

Supplies: Thin Prep Media with Spatula and Brush Kit (T434)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm water to warm and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire exocervix while maintaining tight contact with exocervical surface.

2. Rinse spatulas quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate one-quarter or one-half turn in on direction. **Do not over-rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

9. Place labels on the vial and on the bag.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	THIN PREP
	Refrigerated	42 days	THIN PREP

CPT Code Information:

88142

88141 (if appropriate)

STHPV 70335 ThinPrep Screen with Human Papillomavirus (HPV) Reflex, Varies

Specimen Requirements:

Patient Preparation: For optimal interpretation, Papanicolaou smears should be collected near the middle of the menstrual cycle. No douching, lubricant use, and sexual intercourse for 24 hours prior to specimen collection.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap Test, regardless of the volume of the aliquot (maximum aliquot volume: 4 mL).

Submit only 1 of the following specimens:

Supplies: Thin Prep Media with Broom Kit (T056)

Specimen Type: Cervical

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm water to warm and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

7. Place labels on the vial and on the bag.

Supplies: Thin Prep Media with Spatula and Brush Kit (T434)

Specimen Type: Ectocervix and endocervix

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm water to warm and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire exocervix while maintaining tight contact with exocervical surface.

2. Rinse spatulas quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate one quarter or one half turn in 1 direction. **Do not over rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

9. Place labels on the vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	THIN PREP
	Refrigerated	42 days	THIN PREP

CPT Code Information:

G0123

88142

88141 (if appropriate)

TPRPS ThinPrep Screen, Varies
70332

Specimen Requirements:

Patient Preparation: For optimal interpretation, Pap smears should be collected near the middle of the menstrual cycle. Avoid douching, lubricant use, and sexual intercourse for 24 hours prior to specimen collection.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap test, regardless of the volume of the aliquot (maximum aliquot volume: 4 mL).

Submit only 1 of the following specimens:

Specimen Type: Cervical

Supplies: Thin Prep Media with Broom Kit (T056)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm to warm water and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

7. Place labels on the vial and on the bag.

Specimen Type: Ectocervix and endocervix

Supplies: Thin Prep Media with Spatula and Brush Kit (T434)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm to warm water and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire exocervix while maintaining tight contact with exocervical surface.

2. Rinse spatulas quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate one-quarter or one-half turn in one direction. **Do not over rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

9. Place labels on the vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	THIN PREP
	Refrigerated	42 days	THIN PREP

CPT Code Information:

G0123 (Government Payers)

88142

88141-TPSPC (if appropriate)

DTPCO 70338 ThinPrep with Human Papillomavirus (HPV) Co-Test-Diagnostic, Varies

Specimen Requirements:

Patient Preparation: For optimal interpretation, Papanicolaou smears should be collected near the middle of the menstrual cycle. No douching, lubricant use, and sexual intercourse for 24 hours prior to specimen collection.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap Test, regardless of the volume of the aliquot (maximum aliquot volume: 4 mL).

Submit only 1 of the following specimens:

Specimen Type: Cervical

Supplies: ThinPrep Media with Broom Kit (T056)

Collection Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm water to warm and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

7. Place labels on the vial and on the bag.

Specimen Type: Ectocervix and endocervix

Supplies: ThinPrep Media with Spatula and Brush Kit (T434)

Collection Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm water to warm and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire exocervix while maintaining tight contact with exocervical surface.

2. Rinse spatulas quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate one-quarter or one-half turn in 1 direction. **Do not over rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

9. Place labels on the vial and on the bag.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	THIN PREP
	Refrigerated	42 days	THIN PREP

CPT Code Information:

88142

G0123 (for government payers)

87626 (if appropriate)

87626 (if appropriate)

88141 (if appropriate)

CTPCO 616073 ThinPrep with Human Papillomavirus (HPV) Co-Test-Screen with p16/Ki67 Dual Stain Reflex, Varies

Specimen Requirements:

Patient Preparation: For optimal interpretation, Papanicolaou (Pap) smears should be collected near the middle of the menstrual cycle. No douching, lubricant use, or sexual intercourse for 24 hours before specimen collection.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap test, regardless of the volume of the aliquot (maximum aliquot volume: 4 mL).

Submit only 1 of the following specimens:

Specimen Type: Cervical

Supplies: Thin Prep Media with Broom Kit (T056)

Container/Tube: ThinPrep/PreservCyt vial

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm water to warm and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the broom collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they tend to leak during transport.

7. Place labels on the vial and on the bag.

Specimen Type: Ectocervix and endocervix

Supplies: Thin Prep Media with Spatula and Brush Kit (T434)

Container/Tube: ThinPrep/PreservCyt vial

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm water to warm and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire ectocervix while maintaining tight contact with ectocervical surface.

2. Rinse spatula as quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate one-quarter or one-half turn in 1 direction. **Do not over-rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they tend to leak during transport.

9. Place labels on the vial and bag.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Ambient (preferred)	42 days	THIN PREP
	Refrigerated	42 days	THIN PREP

CPT Code Information:

88142

G0123 (Government payers)

G0124 (Government payers, if appropriate)

88141 -TPSPC (if appropriate)

88344 -CINTC/CINPC (if appropriate)

STPCO 70337 ThinPrep with Human Papillomavirus (HPV) Co-Test-Screen, Varies

Specimen Requirements:

Patient Preparation: For optimal interpretation, Papanicolaou smears should be collected near the middle of the menstrual cycle. No douching, lubricant use, and sexual intercourse for 24 hours prior to specimen collection.

Only 1 aliquot may be removed from PreservCyt sample vial prior to performing the ThinPrep Pap Test, regardless of the volume of the aliquot (maximum aliquot volume: 4 mL).

Submit only 1 of the following specimens:

Specimen Type: Cervical

Supplies: Thin Prep Media with Broom Kit (T056)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain adequate sampling from cervix using a broom-like collection device. If desired, use lukewarm water to warm and lubricate the speculum. Insert the central bristles of the broom into the endocervical canal deep enough to allow the shorter bristles to fully contact the ectocervix. Push gently and rotate the broom in a clockwise direction 5 times.

2. Rinse the broom as quickly as possible into the PreservCyt solution vial by pushing broom into bottom of vial 10 times, forcing the bristles apart.

3. As a final step, swirl broom vigorously to further release material. Discard the collection device.

4. Tighten cap on vial so that the torque line on the cap passes the torque line on the vial.

5. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

6. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

7. Place labels on the vial and on the bag.

Specimen Type: Ectocervix and endocervix

Supplies: Thin Prep Media with Spatula and Brush Kit (T434)

Container/Tube: ThinPrep

Specimen Volume: 16 mL

Collection Instructions:

1. Obtain an adequate sampling from the ectocervix using a plastic spatula. If desired, use lukewarm water to warm and lubricate the speculum. Select contoured end of plastic spatula and rotate it 360 degrees around the entire exocervix while maintaining tight contact with exocervical surface.

2. Rinse spatulas quickly as possible into the PreservCyt solution vial by swirling spatula vigorously in vial 10 times. Discard the spatula.

3. Next, obtain an adequate specimen from endocervix using an endocervical brush device. Insert the brush into the cervix until only the bottommost fibers are exposed. Slowly rotate one-quarter or one-half turn in 1 direction. **Do not over rotate.**

4. Rinse the brush as quickly as possible in the PreservCyt solution by rotating the device in the solution 10 times while pushing against the PreservCyt vial wall.

5. Swirl brush vigorously as final step to further release material. Discard the brush.

6. Tighten the cap so that the torque line on the cap passes the torque line on the vial.

7. **Specimen vial must be labeled with a minimum of 2 unique identifiers** (patient's name and medical record number or date of birth).

8. Bag ThinPrep specimens individually as they have a tendency to leak during transport.

9. Place labels on the vial and on the bag.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container

Varies	Ambient (preferred)	42 days	THIN PREP
	Refrigerated	42 days	THIN PREP

CPT Code Information:

88142

G0123 (for government payers)

87626 (if appropriate)

87626 (if appropriate)

88141 (if appropriate)

FFTIO
57708

Thiocyanate, Serum

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 1 mL sodium heparin plasma refrigerated in plastic vial.

Serum

Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 1 mL of serum refrigerated in plastic vial.

Specimen Minimum Volume: 0.50 mL Does not allow for repeat testing

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

84430

THIO
65381

Thiopurine Metabolites, Whole Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not** aliquot, centrifuge, or freeze.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	8 days
	Ambient	24 hours

CPT Code Information: 80299

TPNUQ
610062 Thiopurine Methyltransferase (TPMT) and Nudix Hydrolase (NUDT15) Genotyping, Varies

Specimen Requirements:

Multiple genotype tests can be performed on a single specimen after a single extraction. See Multiple Genotype Test List for a list of tests that can be ordered together.

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

0034U

TPMT3 65188 Thiopurine Methyltransferase Activity Profile, Erythrocytes

Specimen Requirements:

Patient Preparation: Thiopurine methyltransferase (TPMT) enzyme activity can be inhibited by several drugs and may contribute to falsely low results. Patients should abstain from the following drugs for at least 48 hours prior to TPMT testing: naproxen (Aleve), ibuprofen (Advil, Motrin), ketoprofen (Orudis), furosemide (Lasix), sulfasalazine (Azulfidine), mesalamine (Asacol), olsalazine (Dipentum), mefenamic acid (Ponstel), trimethoprim (Proloprim), methotrexate, thiazide diuretics, and benzoic acid inhibitors.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Green top (sodium or lithium heparin), dark blue top (metal free sodium heparin), or non-centrifuged plasma gel tubes

Specimen Volume: 5 mL

Collection Instructions: Send whole blood specimen. **Do not centrifuge**.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

84433

FSULU
75890

Thiosulfate, Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 4 mL

Collection Instructions:

1. Collect 4 mL random urine specimen in a plastic preservative-free urine container.

2. Send refrigerated.

Specimen Minimum Volume: 1.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days

	Frozen	14 days

CPT Code Information:

82542

82570

81002 (if appropriate)

TTSC 602184 Thrombin Time (Bovine), Plasma

Specimen Requirements:

Specimen Type: Platelet-poor plasma

Patient Preparation:

Fasting: 8 hours, preferred but not required

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 1 mL Platelet-poor plasma

Collection Instructions:

1. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

2. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

3. Aliquot plasma into a separate plastic vial leaving 0.25 mL in the bottom of centrifuged vial.

4. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally at -40 degrees C or below.

Additional Information:

1. A double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: Platelet-poor plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85670

FATCX
75958

Thrombin-Antithrombin Complex

Specimen Requirements:

Collection Container/Tube: Blue top (3.2% buffered sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Plasma

Collection Instructions:

1. Do not draw from an arm with a heparin lock or heparinized catheter.

2. Draw blood in a blue-top tube containing 3.2% buffered sodium citrate.

3. Mix immediately by gently inverting at least 6 times.

4. Centrifuge, aliquot plasma into a plastic vial, being careful not to disturb the cells, and freeze immediately.

5. Send frozen.

Note: **Critical frozen. Separate specimens must be submitted when multiple tests are ordered.**

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	28 days

CPT Code Information:

83520

AATHI
603184

Thrombophilia Profile Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26 Special Coagulation Interpretation

AATHR
603304

Thrombophilia Profile, Plasma and Whole Blood

Specimen Requirements:

Both blood and plasma are required.

Patient Preparation:

1. Specimen **must be** collected prior to initiation of anticoagulants and thrombolytic therapy.

2. Patient **should not** be receiving warfarin, heparin, direct thrombin inhibitors (argatroban, dabigatran), or direct factor Xa inhibitors (apixaban, rivaroxaban, and edoxaban).

a. If medically feasible, for 4 to 6 hours before specimen collection, **do not** administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, **do not** administer subcutaneous heparin or warfarin**.**

3. It is recommended that specimens be collected pretransfusion. If patient has been transfused, **a specimen should not be collected for 48 hours.**

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD), light-blue top (3.2% sodium citrate)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Type: Platelet-poor plasma

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 5 mL Platelet-poor plasma in 5 plastic vials; each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 5 separate plastic vials leaving 0.25 mL in the bottom of centrifuged vial.

5. Immediately freeze plasma (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information: Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

Specimen Minimum Volume: Platelet-poor plasma: 5 mL total, 5 plastic vials each containing 1 mL, Whole blood: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days
Whole blood	Ambient (preferred)	14 days
	Frozen	14 days
	Refrigerated	14 days

CPT Code Information:

81240-F2 PTNT

85300-ATTF

85303-CFX

85306-PSF

85307-APCRV

85379-DIMER

85384-CLFIB

85390-26-AATHI

85610-PTSC

85613-DRV1

85670-TTSC

85730-APTSC

81241-F5 (coagulation factor V) (eg, hereditary hypercoagulability) gene analysis, Leiden variant (if appropriate)

85210-Factor II (if appropriate)

85220-Factor V (if appropriate)

85230-Factor VII (if appropriate)

85240-Factor VIII (if appropriate)

85250-Factor IX (if appropriate)

85260-Factor X (if appropriate)

85270-Factor XI (if appropriate)

85280-Factor XII (if appropriate)

85301-Antithrombin antigen (if appropriate)

85302-Protein C antigen (if appropriate)

85305-Protein S antigen, total (if appropriate)

85306-Protein S activity (if appropriate)

85366-Soluble fibrin monomer (if appropriate)

85385-PT-Fibrinogen (if appropriate)

85597-Platelet neutralization for lupus inhibitor (if appropriate)

85598-Hex LA (if appropriate)

85611-PT mix 1:1 (if appropriate)

85613-DRVVT mix (if appropriate)

85613-DRVVT confirmation (if appropriate)

85635-Reptilase (if appropriate)

85732 - APTT Mix 1:1 (if appropriate)

FFTPO 57822 Thrombopoietin (TPO)

Specimen Requirements:

Specimen Type: Serum
Container/Tube: Red or SST
Specimen Volume: 1 mL
Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 1 mL of serum frozen in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	30 days

CPT Code Information:

83520

GNTHR 619271 Thrombosis Disorders, Comprehensive Gene Panel, Next-Generation Sequencing, Varies

Specimen Requirements:

Specimen Type: Whole blood

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

THSIF 605244 Thrombospondin Type 1 Domain Containing 7A (THSD7A), Immunofluorescence

Specimen Requirements:

Specimen Type: Kidney tissue

Preferred: 2 Unstained positively charged glass slide (25- x 75- x 1-mm) per test ordered; paraffin sections 3 to 4-microns thick

Acceptable: Formalin-fixed, paraffin-embedded (FFPE) kidney tissue block

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

88346-Primary IF

88350-If additional IF

THSD7 607079 Thrombospondin Type-1 Domain-Containing 7A Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container: Plastic vial

Specimen Volume: 1 mL

Collection Information: Centrifuge and aliquot serum into plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	8 hours

CPT Code Information:

86255

THYM 82606 Thyme, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TGAB
84382

Thyroglobulin Antibody, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Red top (gel tubes/**SST are not acceptable)**

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86800

TGABI
610071

Thyroglobulin Antibody, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see IETG / Interference Evaluation Heterophile, Thyroglobulin Tumor Marker, Serum.

Patient Preparation: For 12 hours before specimen collection, patient should not take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: None (serum gel/SST are **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86800

THYR
70565

Thyroglobulin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

TGMS
62749

Thyroglobulin Mass Spectrometry, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.25 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	416 days
	Ambient	72 hours

CPT Code Information:

84432

TGII 610072 Thyroglobulin, Interference Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	90 days
	Ambient	7 days

HTGR 62936 Thyroglobulin, Tumor Marker Reflex, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: None (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

86800

84432 (if appropriate)

TFNAB 607714 Thyroglobulin, Tumor Marker, Fine-Needle Aspiration Biopsy Needle Wash

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Collection Container/Tube: Plain, plastic, screw-top tube

Specimen Volume: 1 to 1.5 mL

Collection Instructions:

1. Needle wash specimens for analysis should be collected in conjunction with cytology specimens.

2. Have saline available prior to start of procedure. **Saline is the only acceptable solution for needle washings.**

3. After each fine-needle aspiration biopsy (FNAB) has been collected and the material in the needle has been expelled onto a slide for cytologic analysis, attach the used FNAB needle to an empty syringe.

4. Withdraw between 0.10 mL and 0.25 mL of saline up through the needle until the saline starts to fill the hub of the needle or end of the syringe.

5. Expel this fluid back through the needle into a separate plastic screw-top tube. This is the needle washing used for analysis.

6. Repeat steps 2 through 4 for each needle pass of the same biopsied site and empty into the same tube, accumulating a total of 0.5 mL to 1.5 mL of fluid to send to the laboratory. (If more than 1 site is biopsied, see Additional Information)

7. Inspect specimen for visible blood or tissue contamination:

a. If bloody, centrifuge specimen and transfer supernatant to a new plastic aliquot tube (5-mL standard tube) prior to sending to laboratory. The supernatant, not the cellular material, is used for analysis.

b. If specimen is clear, centrifugation is not necessary.

Additional Information:

1. If more than 1 site is biopsied, each washing material should be submitted on a separate tube and under a different order number.

2. A minimum of 0.5 mL is required for testing; however, the total collection volume should not exceed 1.5 mL. Specimen volumes outside these parameters may be rejected.

3. Do not send a saline control. This test has been validated to rule-out saline matrix effect.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Fine Needle Wash	Frozen (preferred)	90 days
	Refrigerated	14 days
	Ambient	7 days

CPT Code Information:

84432

HTG2 Thyroglobulin, Tumor Marker, Serum

62800

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84432

86800

TGQN
610070

Thyroglobulin, Tumor Marker, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see IETG / Interference Evaluation Heterophile, Thyroglobulin Tumor Marker, Serum.

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7)

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: None (serum gel/SST are **not** acceptable)

Submission Container/Tube: Plastic vial

Specimen Volume: 2.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	90 days

	Ambient	7 days

CPT Code Information:

84432

TAB 82041 Thyroid Autoantibodies Profile, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86376-Thyroperoxidase antibody

86800-Thyroglobulin antibody

THCSM 622850 Thyroid Function Cascade, Monitor (without Thyroid Peroxidase Antibody), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

84443

84439-(if appropriate)

84480-(if appropriate)

THSCM
83633

Thyroid Function Cascade, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

84443

84439 (if appropriate)

84480 (if appropriate)

86376 (if appropriate)

TTF8G 70575 Thyroid Transcription Factor (8G7G3/1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TTFSP 70576 Thyroid Transcription Factor (SPT24) Immunostain, Technical Component Only

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostains Technical Only Envelope

Preferred: 2 Unstained positively charged glass slide (25- x 75- x 1-mm) per test ordered; sections 4-microns thick.

Acceptable: Formalin-fixed, paraffin-embedded tissue block

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TTFK5 70489 Thyroid Transcription Factor 1 (TTF1) (SPT24) and Keratin 5 (KRT5) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344-TC

TSH
70574

Thyroid-Stimulating Hormone (TSH), Beta Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However,

due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

STSH 8939 Thyroid-Stimulating Hormone-Sensitive (s-TSH), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

84443

FFTSI
75917

Thyroid-stimulating Immunoglobulin (TSI)

Specimen Requirements:

Collection Container/Tube:

Preferred: Red-top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 0.3 mL Note: This volume does not allow for repeat testing

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

84445

TSI
8634

Thyroid-Stimulating Immunoglobulin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	60 days
	Refrigerated	7 days
	Ambient	24 hours

CPT Code Information:

84445

TPO
81765

Thyroperoxidase Antibodies, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86376

THYRO
81797

Thyrotropin Receptor Antibody, Serum

Specimen Requirements:

Patient Preparation:

1. For 12 hours before specimen collection. patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

2. Patient **should not** be receiving heparin treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days

CPT Code Information:

83520

TBGI
9263

Thyroxine-Binding Globulin (TBG), Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

84442

FGTIA 75019 Tiagabine (Gabitril), Serum

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80199

STICK 602733 Tick-Borne Antibodies, Modified 2-Tier, ELISA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.35 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	10 days
	Frozen	14 days

CPT Code Information:

86618

86666 x 2

86753

86617 x 2 (if appropriate)

TICKS
83265

Tick-Borne Disease Antibodies Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	10 days
	Frozen	14 days

CPT Code Information:

86618

86666 x 2

86753

86617 x 2-Lyme disease Western blot (if appropriate)

TIKLB
618311

Tick-Borne Panel, Molecular Detection, PCR, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube (preferred).

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798 x4

87469

87468

87484

87478

87999 (if appropriate for government payers)

BRBST 620148 Tickborne Bacterial, PCR and Sequencing, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: If not submitting in original vial, mix well before transferring to a sterile vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

87801-Broad Range Bacterial PCR and Sequencing

87798-Specimen Identification by PCR (if appropriate)

TILAP 619510 Tilapia, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TIMG
82891

Timothy Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FFTIB
91137

Tin, Blood

Specimen Requirements:

Draw blood in a royal blue-top (metal free EDTA) tube. Send 2 mL metal free EDTA whole blood refrigerated.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA - Metal Free (ERB)	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

83018

FFTIN
91101

Tin, Serum

Specimen Requirements:

Submit only 1 of the following specimens:

Serum

Draw blood in a metal free tube(s), **serum gel tube is not acceptable.** Spin down and send 2 mL of metal free serum refrigerated in a plastic vial.

Plasma

Draw blood in an EDTA metal-free royal blue-top tube(s), **plasma gel tube is not acceptable.** Spin down and send 2 mL metal-free EDTA plasma refrigerated in a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days

	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

83018

TSTGP
83671

Tissue Transglutaminase Antibodies, IgA and IgG Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86364 x 2

TTGA
82587

Tissue Transglutaminase Antibody, IgA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86364

TTGG
83660

Tissue Transglutaminase Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86364

TIWB
614612

Titanium, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Collection Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381)

Specimen Volume: 0.55 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original collection tube. **Do not aliquot.**

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83018

TIS
89367

Titanium, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free Specimen Vial (T173)

-Metal Free B-D Tube (No Additive), 6 mL (T184)

Collection Container/Tube: Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Metal-free, screw-capped, polypropylene vial

Specimen Volume: 0.55 mL Serum

Collection Instructions:

1. Allow specimen to clot for 30 minutes; then centrifuge the specimen to separate serum from the cellular fraction.

2. Remove the stopper. Carefully pour specimen into a metal-free, polypropylene vial, avoiding transfer of the cellular components of blood. **Do not** insert a pipet into the serum to accomplish transfer, and **do not** ream the specimen with a wooden stick to assist with serum transfer.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: Serum: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	28 days	METAL FREE
	Frozen	28 days	METAL FREE
	Ambient	7 days	METAL FREE

CPT Code Information:

83018

TOBPA Tobramycin, Peak, Serum
37063

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80200

TOBRA
37065

Tobramycin, Random, Serum

Specimen Requirements:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and aliquoted within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80200

TOBTA
37064

Tobramycin, Trough, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood no more than 30 minutes before next scheduled dose.

2. Serum gel tubes should be centrifuged within 2 hours of collection.

3. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80200

FHIPP
91121

Toluene as Hippuric Acid, Occupational Exposure, Urine

Specimen Requirements:

10 mL aliquot of random or spot urine collected at end of shift. Send specimen refrigerated. Tubes should be filled to prevent loss of volatile compound into headspace.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	180 days

CPT Code Information:

83921

82570

FFTLB
91141

Toluene, Occupational Exposure, Blood

Specimen Requirements:

Draw blood in 2 green top (sodium heparin) tubes. Tubes should be filled to prevent loss of volatile compound into headspace. Send 20 mL of sodium heparin whole blood refrigerated.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
WB Sodium Heparin	Refrigerated (preferred)	14 days
	Frozen	365 days

CPT Code Information:

84600

FMATG
57628

Tomato IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

TOMA
82695

Tomato, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TOPI
81546

Topiramate, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80201

TRCHG 800270 ToRCH Profile IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86644-CMV

86695-Herpes simplex, type 1

86696-Herpes simplex, type 2

86762-Rubella

86777-*Toxoplasma*

TIBC
2501

Total Iron Binding Capacity, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see SFEC / Iron and Total Iron Binding Capacity, Serum

Patient Preparation:

1. Fasting: 8 hours, preferred but not required

2. For 24 hours before collection, patient **should not** take iron-containing supplements.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood before noon (preferred).

2. Within 2 hours of collection, serum gel tubes should be centrifuged.

3. Within 2 hours of collection, red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	365 days

CPT Code Information:

83550

TOXCG
616734

Toxocara Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	5 days

CPT Code Information:

86682

FGGMC
75519

Toxoplasma gondii Antibodies (IgG, IgM), ELISA, CSF

Specimen Requirements:

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions:

1. Submit 1 mL of spinal fluid (CSF) in a sterile, leak-proof container.

2. Send refrigerate.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	7 days

CPT Code Information:

86777-IgG

86778-IgM

TOXGP
800077

Toxoplasma gondii Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86777

TXPG 621838 Toxoplasma gondii Antibody, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	90 days
	Ambient	72 hours

CPT Code Information:

86777

TXMGP 802127 Toxoplasma gondii Antibody, IgM and IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Aliquot tube

Specimen Volume: 1.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86778-Toxoplasma IgM

86777-Toxoplasma IgG

TXPAB
621836

Toxoplasma gondii Antibody, IgM and IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	21 days
	Frozen	90 days
	Ambient	72 hours

CPT Code Information:

86778-Toxoplasma IgM

86777-Toxoplasma IgG

TXM
802126

Toxoplasma gondii Antibody, IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.8 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86778

TXPM
621837

Toxoplasma gondii Antibody, IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	90 days
	Ambient	72 hours

CPT Code Information:

86778

TOXB 62977 Toxoplasma gondii, Molecular Detection, PCR, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

PTOX 81795 Toxoplasma gondii, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Amniotic fluid

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Spinal fluid

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: 12 x 75-mm screw cap vial

Acceptable: Sterile vial

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Type: Fresh tissue

Supplies: M4-RT (T605)

Container/Tube:

Preferred: Multi-microbe medium (eg, M4-RT)

Acceptable: Sterile container with 1 to 2 mL of sterile saline

Specimen Volume: Entire collection

Collection Instructions: Submit only fresh tissue in a sterile container containing 1 mL to 2 mL of sterile saline or multi-microbe medium (M4-RT, M4, or M5)

Specimen Type: Ocular fluid

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container: 12 x 75-mm screw cap vial **(swabs are NOT acceptable)**

Specimen Volume: 0.3 mL

Collection Instructions:

1. Aliquot collected fluid into screw-cap vial. **Do not submit ocular fluid in syringe.**

2. **Do not centrifuge or dilute the specimen.**

Specimen Minimum Volume: Amniotic fluid, Ocular fluid, Spinal fluid: 0.3 mL; Tissue: 2 x 2 mm biopsy

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

TOXO
70569

Toxoplasma Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

CSP53 607599 TP53 Gene Somatic Mutation Pre-Analysis Cell Sorting, Varies (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see P53CA / Hematologic Neoplasms, *TP53* Somatic Mutation, DNA Sequencing Exons 4-9, Varies.

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	10 days
	Refrigerated	10 days

CPT Code Information:

88184-Flow cytometry, first cell surface, cytoplasmic or nuclear marker

88185 x 4-Each additional marker

TP53 TP53 Mutation Analysis, Next-Generation Sequencing, Tumor

616496

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81351

TRAM 62595 Tramadol and Metabolite, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

G0480

80373 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

TRAMS
623268

Tramadol and Metabolite, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

G0480

80373 (if appropriate for select payers)

TFE3I
70564

Transcription Factor E3 (TFE3) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	

	Refrigerated

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

TRSF
34623

Transferrin, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For serum gel tubes, aliquot serum into a plastic vial prior to shipment.

3. For red-top tubes, aliquot the serum into a plastic vial immediately after centrifuging.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	7 days

CPT Code Information:

84466

FFTRZ
75024

Trazodone (Desyrel)

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 1 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80338

FHEAV
57949

Tree of Heaven (Ailanthus spp) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

TREE1 Tree Panel #1, Serum
81886

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TREE3 Tree Panel #3, Serum
81704

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TREE4
81705

Tree Panel #4, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TRCNG
616733

Trichinella Antibody, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	30 days
	Refrigerated	5 days

CPT Code Information:

86784

FFTRU 91099 Trichloroacetic Acid, Urine

Specimen Requirements:

Submit a 10 mL aliquot from a random or spot urine collected at end of shift, end of exposure, or end of workweek. Send specimen refrigerated.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	180 days

CPT Code Information:

83921-Organic acid, single, quantitative

82570-Creatine, other source

TRVI 82853 Trichoderma viride, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MTRNA 61756 Trichomonas vaginalis, Nucleic Acid Amplification, Varies

Specimen Requirements:

This test should only be performed in men.

Submit only 1 of the following specimens:

Specimen Type: Post-prostatic massage fluid (VBIII)

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima Urine Specimen Transport Tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should void a small amount of urine prior to prostatic massage. Pre-massage urine can be discarded or submitted for other testing as applicable.

3. Patient then ceases voiding and a prostatic massage is performed by the urologist or other health care professional.

4. Collect post-massage fluid into a sterile, plastic, preservative-free container.

5. Transfer 2 mL of post-massage fluid specimen into the Aptima urine specimen transport tube using the disposable pipette provided within 24 hours of collection. The correct volume of fluid has been added when the fluid level is between the black fill lines on the Aptima urine transport tube.

Specimen Type: Urethral (male only)

Supplies: Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: Swab

Collection Instructions:

1. **Urethral specimens must be collected using an Aptima Unisex Swab Collection kit**.

2. Patient should not have urinated for at least 1 hour prior to collection.

3. With a rotating movement, insert swab (blue shaft) 2 to 4 cm into urethra.

4. Once inserted, rotate blue swab gently at least 1 full rotation using sufficient pressure to ensure swab comes into contact with all urethral surfaces. Allow blue swab to remain inserted for 2 to 3 seconds.

5. Place blue swab in the Aptima transport tube provided in collection kit.

6. Snap off blue swab at score line so it fits into closed tube.

7. Cap tube securely and label tube with patient's entire name and collection date and time.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)	30 days	APTIMA VIAL
	Ambient	30 days	APTIMA VIAL

CPT Code Information:

87661

TVRNA 61755 Trichomonas vaginalis, Nucleic Acid Amplification, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Endocervix/Cervix

Supplies: Aptima Unisex Swab Collection Kit (T583)

Container/Tube: Aptima Unisex Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. **Specimens must be collected using the Aptima Unisex Swab Collection Kit**.

2. Use cleaning swab (white shaft) to remove excess mucus from endocervix/cervix.

Discard the cleaning swab.

3. Insert second swab (blue shaft) 1 to 1.5 cm into endocervical canal, and rotate swab gently for 30 seconds. Avoid touching vaginal wall when removing swab.

4. Place blue swab into Aptima transport tube provided in collection kit.

5. Snap off blue swab at score line so it fits into closed tube.

7. Cap tube securely and label tube with patient's entire name and collection date and time.

8. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred), transport within 60 days of collection.

Specimen Type: Vaginal

Supplies: Aptima Multitest Swab Collection Kit (T584)

Container/Tube: Aptima Multitest Swab

Specimen Volume: 1 Swab

Collection Instructions:

1. **Specimens must be collected using the Aptima Multitest Swab Collection Kit**.

2. Insert swab (pink shaft) about 5 cm past introitus and rotate gently for 30 seconds.

3. Place pink swab into Aptima transport tube provided in collection kit.

4. Snap off pink swab at score line so it fits into closed tube.

5. Cap tube securely and label tube with patient's entire name and collection date and time.

6. Maintain swab container at 2 to 30 degrees C (refrigerate temperature is preferred), transport within 60 days of collection.

Specimen Type: ThinPrep Specimen (Endocervix/Cervix)

Supplies: Aptima Specimen Transfer Kit (T652)

Container/Tube: Aptima specimen transfer tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect ThinPrep sample as per normal collection process.

2. **ThinPrep specimen must be aliquoted (as outlined below) before it is processed/tested for Papanicolaou testing (Pap smear)**.

3. Vortex ThinPrep/PreservCyt vial 3 to 10 seconds.

4. Within 1 minute of vortexing:

a. Transfer 1 mL of specimen into the Aptima specimen transfer tube using a disposable transfer pipette or a pipette tip containing a filter (aerosol barrier or hydrophobic plug).

b. Process only 1 ThinPrep and transfer tube set at a time.

c. Recap Aptima specimen transfer tube tightly and gently invert 3 times to mix.

4. Label Aptima transfer tube with appropriate label.

5. Use remainder of ThinPrep specimen for Pap testing.

6. Maintain specimen transport tube at 2 to 30 degrees C (refrigerate temperature is preferred), transport within 30 days of collection when stored at 2 to 14 degrees C or within 14 days when stored at 15 to 30 degrees C.

Specimen Type: Urine

Supplies: Aptima Urine Transport Tube (T582)

Container/Tube: Aptima urine specimen transport tube

Specimen Volume: 15 to 20 mL

Collection Instructions:

1. Patient should not have urinated for at least 1 hour prior to specimen collection.

2. Patient should collect first portion of random voided urine (first part of stream) into a sterile, plastic, preservative-free container.

3. Within 24 hours of collection, transfer 2 mL of urine into the Aptima urine specimen transport tube using the disposable pipette provided. The correct volume of urine has been added when the fluid level is between the black fill lines on the Aptima urine transport tube.

4. Maintain Aptima urine specimen transport tube at 2 to 30 degrees C (refrigerate temperature is preferred), transport within 30 days of collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Refrigerated (preferred)		APTIMA VIAL
	Frozen	180 days	APTIMA VIAL
	Ambient		APTIMA VIAL

CPT Code Information:

87661

FFTMV 75554 Trichophyton Mentagrophytes (var interdigitale) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

TCPT
82720

Trichophyton rubrum, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

TRPS1
616890

Trichorhinophalangeal syndrome type 1 (TRPS1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TGLBF
606918

Triglycerides, Body Fluid

Specimen Requirements:

Specimen Type: Body fluid

Preferred Source:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, Jackson Pratt [JP] drain)

-Pericardial Fluid

Acceptable Source: Write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Frozen (preferred)	30 days
	Refrigerated	7 days
	Ambient	24 hours

CPT Code Information:

84478

TRIGC
606879

Triglycerides, CDC, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see LMPP / Lipoprotein Metabolism Profile, Serum.

Patient Preparation:

1. **Fasting: 12 hours, required**

2. Patient **must not** consume any alcohol for 24 hours before specimen collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	60 days

CPT Code Information:

84478

TRIG1
616665

Triglycerides, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.5 mL

Collection Container/Tube: Plastic vial

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	7 days
	Frozen	90 days

CPT Code Information:

84478

TMP
80146

Trimethoprim, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (gel tubes/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum for a peak level should be collected at least 60 minutes after a dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

TRMP
64269

Trimipramine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Collect specimen immediately before next scheduled dose (minimum 12 hours after last dose).

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial. Serum **must be** separated from cells within 2 hours of collection.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

80299

TPI1
607458

Triosephosphate Isomerase Enzyme Activity, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood in original tube. **Do not** aliquot.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82657

TPIC
608424

Triosephosphate Isomerase Enzyme Activity, Blood (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-HAEV1 / Hemolytic Anemia Evaluation, Blood

-EEEV1 / Red Blood Cell (RBC) Enzyme Evaluation, Blood

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	20 days

CPT Code Information:

82657

T46CS
616449

Tripartite Motif-Containing Protein 46 (TRIM46) IgG, Cell Binding Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

T46CC 616450 Tripartite Motif-Containing Protein 46 (TRIM46) IgG, Cell Binding Assay, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

T46TS 616447 Tripartite Motif-Containing Protein 46 (TRIM46) IgG, Tissue Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

T46TC 616448 Tripartite Motif-Containing Protein 46 (TRIM46) IgG, Tissue Immunofluorescence Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

T46IS 616445 Tripartite Motif-Containing Protein 46 (TRIM46) IgG, Tissue Immunofluorescence, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation,

DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

T46IC 616446 Tripartite Motif-Containing Protein 46 (TRIM46) IgG, Tissue Immunofluorescence, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

FFTRP 91774 Trofile Co-Receptor Tropism Assay

Specimen Requirements:

Specimen Type: Plasma

Container/Tube: Lavender-top (EDTA) or plasma preparation tube (PPT)

Specimen Volume: 3 mL

Collection Instructions: Draw blood in either PPT (pearl top) or lavender-top (EDTA) tubes.

Centrifuge specimen within six hours of collection. Remove plasma from cells immediately, and transfer specimen to a screw-capped, plastic vial. Freeze 3 mL of PPT plasma or EDTA plasma immediately, send specimen frozen.

To avoid delays in turnaround time when requesting multiple tests, please submit separate frozen specimens for each test requested.

RECOMMENDED:

1. Patient's most recent viral load

2. Viral load collection date

Note: 1. Intended to use only for patients with viral loads greater than or equal to 1000 copies/mL. For best results, viral loads should be confirmed within two weeks prior to submission for testing at Monogram.

2. Patient samples submitted <30 days apart are considered duplicate and will be cancelled.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	14 days

CPT Code Information:

87999

FFTRO 57159 Trofile DNA Co-Receptor Tropism Assay

Specimen Requirements:

Specimen Type: Whole Blood

Container/Tube: Lavender-top (EDTA)

Specimen Volume: 4 mL

Collection Instructions: Draw 4 mL blood in a lavender-top (EDTA) tube(s), (Do not centrifuge.) Freeze and ship frozen.

To avoid delays in turnaround time when requesting multiple tests, please submit separate frozen specimens for each test requested.

Note: Trofile DNA is recommended for patients with undetectable viral loads.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Frozen	14 days

CPT Code Information:

87999

WHIPB 87974 Tropheryma whipplei, Molecular Detection, PCR, Blood

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Tropheryma whipplei* DNA is unlikely.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube (preferred)

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Ambient	7 days
	Frozen	7 days

CPT Code Information:

87798

TWRP
80909

Tropheryma whipplei, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Tropheryma whipplei* DNA is unlikely.

Submit only 1 of the following specimens:

Specimen Type: Fresh tissue or biopsy

Sources: Small intestine (duodenum, ileum, or jejunum), lymph node, bone, joint, synovial, liver, pancreas, spleen, lung, heart valve (and other heart tissues), or brain

Container/Tube: Sterile container

Specimen Volume: Entire collection or 5 mm(3) - approximately the size of a pencil eraser

Collection Instructions:

1. Collect fresh tissue specimen.

2. Submit tissue only, do not add fluid to tissue.

3. Refrigerate or freeze specimen.

Specimen Stability Information: Refrigerated (preferred) 7 days /Frozen 7 days

Preferred Paraffin-embedded tissue block:

Supplies: Tissue Block Container (T553)

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Small intestine (duodenum, ileum, or jejunum), lymph node, bone, joint, synovial, liver, pancreas, spleen, lung, heart valve (and other heart tissues), or brain

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block to be cut and returned.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Acceptable Paraffin-embedded tissue block:

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Small intestine (duodenum, ileum, or jejunum), lymph node, bone, joint, synovial, liver, pancreas, spleen, lung, heart valve (and other heart tissues), or brain

Container/Tube: Sterile container for each individual cut section (scroll).

Collection Instructions: Perform microtomy and prepare five separate 10-micron sections. **Each section (scroll) must be placed in a separate sterile container for submission.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Fluid

Sources: Cerebrospinal or ocular (eg, vitreous humor)

Container/Tube: Sterile vial

Specimen Volume: 0.5 mL

Specimen Stability Information: Refrigerated (preferred) 7 days/Frozen 7 days

Specimen Type: Synovial fluid

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Pink top (EDTA), royal blue top (EDTA), sterile vial containing EDTA-derived aliquot, red clot tube (no anticoagulant), or sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Send fluid specimen in original tube (preferred).

Specimen Stability Information: Refrigerated (preferred) 7 days /Frozen 7 days

Specimen Minimum Volume: Fresh tissue or biopsy: 5 mm(3); Paraffin-embedded tissue block: Two 10-micron sections; All other specimens: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87798

TROP2
621064

Trophoblast Antigen 2 (TROP2) Immunostain, Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

TRK
603300

Tropomyosin Receptor Kinase (TRK) Immunostain, Technical Component Only

Specimen Requirements:

Supplies: Immunostain Technical Only Envelope (T693)

Specimen Type: Tissue

Container/Tube: Immunostains Technical Only Envelope

Preferred: 2 Unstained positively charged glass slide (25- x 75- x 1-mm) per test ordered; sections 4-microns thick.

Acceptable: Formalin-fixed, paraffin-embedded tissue block

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

HSTNI
614422

Troponin I, High Sensitivity, Plasma

Specimen Requirements:

Collection Container/Tube:

Preferred: Mint green top (lithium heparin gel)

Acceptable: Green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Plasma gel tube should be centrifuged within 2 hours of collection.

2. Green-top tube should be centrifuged, and the plasma aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	180 days
	Refrigerated	48 hours
	Ambient	4 hours

CPT Code Information:

84484

TRPS
65832

Troponin T, 5th Generation, Plasma

Specimen Requirements:

Collection Container/Tube:

Preferred: Light-green top (lithium heparin gel)

Acceptable: Green top (lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Within 2 hours of collection, lithium heparin gel tubes should be centrifuged.

2. Within 2 hours of collection, lithium heparin tubes should be centrifuged, and the plasma aliquoted into a plastic vial .

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Li Heparin	Frozen (preferred)	365 days
	Ambient	24 hours
	Refrigerated	24 hours

CPT Code Information:

84484

TROT
82788

Trout, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

CRUZI
621062

Trypanosoma cruzi (Chagas) Antibody Panel, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86753 x2

CHAGL 622447 Trypanosoma cruzi IgG, Lateral Flow Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see CRUZI / *Trypanosoma cruzi* (Chagas) Antibody Panel, Serum.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.2 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.10 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86753

CHAGS 615244 Trypanosoma cruzi Total Antibody, Enzyme-Linked Immunosorbent Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see CRUZI / *Trypanosoma cruzi* (Chagas) Antibody Panel, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

CPT Code Information:

86753

CHAGI 622448

Trypanosoma cruzi, Technical Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see CRUZI / *Trypanosoma cruzi* (Chagas) Antibody Panel, Serum.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen	14 days

TRYPN
70572

Trypsin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TRPTS 70573 Tryptase Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TRYPA 32283 Tryptase, Autopsy, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Additional Information: Tryptase degenerates very quickly when left in the presence of red blood cells.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	14 days
	Refrigerated	7 days

CPT Code Information:

83520

TRYPT
81608

Tryptase, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial

Additional Information: Tryptase degenerates very quickly when left in the presence of red blood cells.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Frozen (preferred)	14 days
	Refrigerated	7 days

CPT Code Information:

83520

TRYPP Tryptophan, Plasma

82955

Specimen Requirements:

Patient Preparation:

Fasting: 4 hours, required; Infants should have specimen collected before next feeding (2-3 hours without total parenteral nutrition if possible)

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Green top (sodium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Collect specimen and place on wet ice. Note: Thrombin-activated tubes **should not be used** for collection.

2. Centrifuge immediately or within 4 hours of collection if the specimen is kept at refrigerated temperature.

3. Being careful to ensure that no buffy coat is transferred, aliquot plasma into a plastic vial and freeze.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	14 days

CPT Code Information:

82131

TRYPU Tryptophan, Random, Urine

83823

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 2 mL

Collection Instructions: Collect a random urine specimen.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	70 days
	Refrigerated	14 days

CPT Code Information:

82131

TTF40
602647

TTF41 (SPT24) + p40 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344-TC

TTRZ
617753

TTR Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin):

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for

Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1year /Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood**

specimen; order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81404

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88235-Amniotic fluid culture (if appropriate)

81265-Maternal cell contamination (if appropriate)

TSCP 616537 Tuberous Sclerosis Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Send: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions T800)

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with